

Discrete mathematics

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Sets

- **Set, element of a set** (notation: $\in$, negation: $\notin$): basic concepts.
- Defining a set: by enumeration, e.g., $\{1, 2, 3\}$,
or with the help of a defining property T concerning the elements of
a given set S in the way $\{x \in S \mid T(x)\}$, e.g.,

$$\{x \in \mathbb{N} \mid 1 \leq x \leq 5\}.$$

- **Emptyset**: the unique set, that doesn't have any element.
Notation: $\emptyset$.
- Notation of the **subset** relation: $\subset$.
 $A \subset B$: all elements of A are elements of B as well.
- Two sets **are equal** or **coincide** if their elements are the same.
Equivalently, if they are each others' subsets:

$$A = B \iff A \subset B \text{ and } B \subset A.$$

Cardinality of sets, power set

Let $S = \{0, 1, 2, 3\}$. How many elements does S have? $\#S = 4$.
Give all possible subsets of S .

$$\mathcal{P}(S) = \{\emptyset, \{0\}, \{1\}, \{2\}, \{3\}, \{0, 1\}, \{0, 2\}, \{0, 3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \\ \{0, 1, 2\}, \{0, 1, 3\}, \{0, 2, 3\}, \{1, 2, 3\}, S\}$$

Definition

The **power set** of a given set S is the set of all subsets of S . Notation: $\mathcal{P}(S)$ or 2^S .

Definition

If a set has a finite number of elements, then this number is called the **cardinality** of the set. Notation for a given set S : $\#S$ or $|S|$.
In this case we say that S is a **finite set**.

Theorem

If S has cardinality of n , then the power set of S has cardinality of 2^n , that is

$$\#(\mathcal{P}(S)) = 2^{\#S}.$$

Fundamental operations on sets

- The **complement** of a set A : $\overline{A}$.

The complement of a set A is the set of all elements that belong to the **universal set** but not to A .

- The **union** of two sets: $A \cup B$.

$A \cup B$: a set containing all elements that are in A **or** in B (possibly both).

- The **intersection** of two sets: $A \cap B$.

$A \cap B$: a set containing all elements that are elements of A **and** B .

- The (set-theoretic) **difference** of two sets: $A \setminus B$.

$A \setminus B$: all elements that are elements of A but not of B .

Operations on sets (cont.)

- The **symmetric difference** (or **disjunctive union**) of two sets: $\triangle$.
 $A \triangle B$: a set containing all elements that are either in A **or** in B (but not in both).

$$A \triangle B = (A \cup B) \setminus (A \cap B) = (A \setminus B) \cup (B \setminus A)$$

E.g., if $A = \{0, 1, 2, 3, 4\}$, $B = \{2, 4, 6, 8, 10\}$ what is $A \triangle B = ?$

- The **Cartesian product** of two sets: $\times$.
 $A \times B$: a set containing all **ordered pairs**, where the first element is from A and the second is from B .

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

E.g., if $A = \{0, 1, 2\}$, $B = \{1, 2\}$ what is $A \times B = ?$

Theorem – De Morgan's laws

If A and B are arbitrary sets, then

$$\overline{(A \cup B)} = \bar{A} \cap \bar{B} \quad \text{and} \quad \overline{(A \cap B)} = \bar{A} \cup \bar{B}.$$

Furthermore, these identities hold for arbitrary number of sets.

Notation

Special sets of numbers:

- $\mathbb{N} = \{1, 2, 3, \dots\}$: the set of natural numbers (to be defined later)
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$: the set of integers
- $\mathbb{Q}$: the set of rational numbers
- $\mathbb{R}$: the set of real numbers
- $\mathbb{C}$: the set of complex numbers (to be defined later)

Quantifiers:

- $\exists$: 'there exists' (existential quantifier)
- $\forall$: 'for all' (universal quantifier)

E.g., $\exists n \in \mathbb{N} : 2n = 6$, but $\nexists n \in \mathbb{N} : 2n = 7$

$\forall m \in \mathbb{N} : m \in \mathbb{Z}$, but $\nexists m \in \mathbb{Z} : m \in \mathbb{N}$

Functions

Function: an association rule, assignment or correspondence $x \mapsto f(x)$

For example:

$$f: \mathbb{R} \rightarrow \mathbb{R}, x \mapsto x^2 - 3x + 2.$$

If the function f accomplishes a correspondence between the set D (the domain of the function) and the set R (the codomain of the function), then we can view the function as pairs $(x, f(x))$, where $x \in D$ and $f(x) \in R$.

$$f: D \rightarrow R, x \mapsto f(x)$$

That is, the function is a subset of the Cartesian product $D \times R$, such that if

$$f: x \mapsto y_1 \quad \text{and} \quad f: x \mapsto y_2,$$

then necessarily $y_1 = y_2$.

Examples of functions

- $x \in \mathbb{R}, x \mapsto f(x) := x^2$
- $x \in \mathbb{R}^+, x \mapsto f(x) := \text{a number with square } x$
Not a function!
- $n \in \mathbb{N}, n \mapsto f(n) := \text{an odd number such that it's a divisor of } n$
Not a function!
- $n \in \mathbb{N}, n \mapsto f(n) := \text{the greatest odd divisor of } n$
Function!

Notation

The meaning of $:=$ is: definition, prescribing a value, 'let it be equal with'

Notation

The meaning of different arrows: $\rightarrow, \mapsto, \Rightarrow, \Leftrightarrow$

Basic functions

- constant: $f(x) = c$
- first order: $f(x) = ax + b, a \neq 0$
- second order: $f(x) = ax^2 + bx + c \quad (a \neq 0)$
factored form: $f(x) = a \cdot \left(x - \frac{-b + \sqrt{b^2 - 4ac}}{2a}\right) \cdot \left(x - \frac{-b - \sqrt{b^2 - 4ac}}{2a}\right)$
- polynomial
- exponential: $f(x) = a^x \quad (a > 0, a \neq 1)$
- logarithmic: $f(x) = \log_a x \quad (a > 0, a \neq 1)$
- trigonometric functions
- absolute value function
- sign function or signum function
- **identical function**: $f(x) = x$

Properties of functions

Let us consider an arbitrary function

$$f: D \rightarrow R, x \mapsto f(x).$$

Definition

The function f is **injective** if $f(a) = f(b)$ implies $a = b$.

That is, in this case the function f assigns a **different** value to each element.

Definition

The function f is **surjective** or **onto** if for every element y in the codomain R there exists an element $x \in D$ such that $f(x) = y$.

That is, f is surjective if all elements of R become an image of an element. (Then $\text{range} = \text{codomain}$.)

Definition

The function f is **bijective** if it is injective and surjective.

Bijection: one-to-one correspondence

Reasoning with mathematical induction

Let us assume that we want to prove a proposition (for example, the relation below) **for all natural numbers**:

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2, \quad \forall n \in \mathbb{N}.$$

Then we can use the following reasoning:

- (1) **We prove** the proposition for $n = 1$. (By trial and error.)
- (2a) **We assume** that the proposition is true for an arbitrary natural number k ,
- (2b) then **we prove** it for the natural number $k + 1$.

(2a): **inductive hypothesis**

If we are able to carry out (1)-(2a)-(2b), then it's proved that the proposition holds for **all natural numbers**.

Step 2 ((2a)+(2b)): **inductive step**:

'property being true for k implies property being true for $k + 1$ '

The set of natural numbers

For the axiomatic introduction of this set we use the so-called Peano axioms.

Definition – Peano axioms

- Ⓟ₁ 1 is a natural number.
- Ⓟ₂ For every natural number n there exists uniquely a **successor** natural number.
- Ⓟ₃ There is no natural number whose successor is 1.
- Ⓟ₄ If two natural numbers have the same successors, then the two natural numbers coincide.
- Ⓟ₅ **Axiom of induction:** if A is a set such that
 - ▶ it contains the natural number 1,
 - ▶ for every element of A its successor is also in A ,then A contains all the natural numbers.

The conditions (P1)–(P5) uniquely determine a set, which is called **the set of natural numbers**. Notation: $\mathbb{N}$.

Remarks on the Peano axioms

(P2) For every natural number n it is possible to provide a 'greater by 1' natural number, which is called the **successor** of n .

$\rightsquigarrow n + 1, S(n)$ (S : successor function)

(P4) If two natural numbers have the same successors, then the two natural numbers coincide.

In other words: the successor function is **injective**.

(P5) **Axiom of induction**: if A is a set such that

- it contains the natural number 1,
- for every element of A its successor is also in A ,

then A contains all the natural numbers.

In other words: A is an **inductive set**. $\Rightarrow \mathbb{N}$ is the smallest inductive set.

Another example for inductive sets: the set of positive numbers ($\mathbb{R}^+$).

The Peano axioms with mathematical formalism

Definition – Peano axioms

Let $\mathbb{N}$ be a set satisfying the following conditions:

- P1 $1 \in \mathbb{N}$
- P2 $\forall n \in \mathbb{N} : \exists S(n) \in \mathbb{N}, S(n) =: n + 1$
or: $\exists S : \mathbb{N} \rightarrow \mathbb{N}$ so-called **successor function**
- P3 $\nexists n \in \mathbb{N} : S(n) = 1$
- P4 $n, m \in \mathbb{N} : S(n) = S(m) \Rightarrow n = m$

P5

$$\left. \begin{array}{l} 1 \in A \\ n \in A \Rightarrow S(n) \in A \end{array} \right\} \Rightarrow \mathbb{N} \subset A$$

Then $\mathbb{N}$ is uniquely determined, and it is called **the set of natural numbers**.

Proof by induction

Based on the definition the elements of $\mathbb{N}$ are:

$$1, S(1), S(S(1)), S(S(S(1))), \dots, S(S(\dots(S(1))\dots)), \dots$$

$$S(1) = 1 + 1 =: 2$$

$$S(S(1)) = S(1) + 1 =: 3$$

The axiom of induction expresses that all the natural numbers can be given with the help of the special natural number 1 and the successor function S . Thus, if we want to prove a proposition (for example, a relation below) for *all natural numbers*, then we can apply the **reasoning of mathematical induction**:

- (1) **We prove** the proposition for $n = 1$. (By trial and error.)
- (2a) **We assume** that the proposition is true for an arbitrary natural number k ,
- (2b) then **we prove** it for the natural number $k + 1$.

(2a): **inductive hypothesis**

Examples for proof by induction

- 1 The sum of the first n natural numbers is $\frac{n(n+1)}{2}$. We can apply induction here. ✓
- 2 $x + \frac{1}{x} \geq 2, \forall x > 0$. We cannot apply induction for this!
- 3 Prove that

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2, \quad \forall n \in \mathbb{N}.$$

- 4 Prove that

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}, \quad n \in \mathbb{N}.$$

Notation

$$\sum_{i=1}^n \text{ sum, } \quad \prod_{i=1}^n \text{ product}$$

Example

Show that

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2, \quad \forall n \in \mathbb{N}.$$

(1) **We prove** the statement for $n = 1$.

$$\left. \begin{array}{l} \text{Left hand side: } 1 \\ \text{Right hand side: } 1^2 = 1. \end{array} \right\} \Rightarrow \text{The statement holds for } n = 1.$$

(2a) **We assume** that the equality holds for an arbitrary natural number k :

$$1 + 3 + 5 + \cdots + (2k - 1) = k^2 \quad (\text{inductive hypothesis}).$$

(2b) **We prove** that the proposition is true for $n = k + 1$.

We have to show that

$$1 + 3 + 5 + \cdots + (2k - 1) + (2(k + 1) - 1) = (k + 1)^2.$$

The calculation (using the inductive hypothesis):

$$\underbrace{1 + 3 + 5 + \cdots + (2k - 1)}_{k^2} + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2.$$

The set of integers

In the set of natural numbers the equation $n + x = m$ is not always solvable. For example:

$$2 + x = 9. \quad x = 7.$$

$$5 + x = 3. \quad x = ?$$

This equation leads us to the notion of **subtraction** and to the set of integers: $\mathbb{Z}$.

- We introduce the set of integers with the help of the already defined set of natural numbers. The precise definition is not discussed here.
- The set of integers is the smallest set which contains the natural numbers and is closed under subtraction.
- When defining $\mathbb{N}$ and $\mathbb{Z}$, we have to define also operations (addition and multiplication) and **ordering**, that is, relations $<$ and $\leq$.

Divisors, divisibility

Let $a, b \in \mathbb{Z}$.

Definition

We say that b is a divisor of a , or a is a multiple of b , or a is divisible by b if there exists $c \in \mathbb{Z}$ such that $a = b \cdot c$.

Notation: $b|a$

Theorem – the properties of divisibility

- ① $\forall a \neq 0, a \in \mathbb{Z} : a|0, 1|a, a|a$
- ② If $a|b$ and $c \in \mathbb{Z}$, then $a|bc$. ($a|b \wedge c \in \mathbb{Z} \Rightarrow a|bc$)
- ③ If $a|b_1$ and $a|b_2$, then $a|(b_1 + b_2)$.
- ④ If $a|b$ and $b|c$, then $a|c$.
- ⑤ If $a|b$ and $b|a$, then $a = \pm b$.

② és ③ $\Rightarrow$ If $a|b_i, i = 1, 2, \dots, n$ and $c_1, c_2, \dots, c_n \in \mathbb{Z}$, then

$$a|(b_1c_1 + b_2c_2 + \dots + b_nc_n).$$

Divisibility rules

$A \in \mathbb{N} \Rightarrow$

$$A = a_n \cdot 10^n + a_{n-1} \cdot 10^{n-1} + \cdots + a_2 \cdot 10^2 + a_1 \cdot 10 + a_0,$$

$$a_i \in \{0, 1, \dots, 9\}, a_n \neq 0.$$

- Divisibility by 2

$$A = (a_n \cdot 10^{n-1} + a_{n-1} \cdot 10^{n-2} + \cdots + a_2 \cdot 10 + a_1) \cdot 10 + a_0$$

$2|10$, thus $2|A$ if and only if $2|a_0$

- Divisibility by 5: A as above

$5|10$, thus $5|A$ if and only if $5|a_0$

Divisibility rules

$A \in \mathbb{N} \Rightarrow$

$$A = a_n \cdot 10^n + a_{n-1} \cdot 10^{n-1} + \cdots + a_2 \cdot 10^2 + a_1 \cdot 10 + a_0,$$

$$a_i \in \{0, 1, \dots, 9\}, a_n \neq 0.$$

- Divisibility by 4: $4 \nmid 10$, but $4 \mid 100$

$$A = (a_n \cdot 10^{n-2} + a_{n-1} \cdot 10^{n-3} + \cdots + a_2) \cdot 100 + a_1 \cdot 10 + a_0$$

$4 \mid 100$, so $4 \mid A$ if and only if $4 \mid (a_1 \cdot 10 + a_0)$

- Divisibility by 25: analogously to 4, since $25 \mid 100$.
- Divisibility by 8: $8 \nmid 100$, however $8 \mid 1000 \Rightarrow$

$$8 \mid A \iff 8 \mid (100a_2 + 10a_1 + a_0)$$

$100a_2 + 10a_1 + a_0$ is the remainder when dividing A by 1000.

Divisibility rules

$A \in \mathbb{N} \Rightarrow$

$$A = a_n \cdot 10^n + a_{n-1} \cdot 10^{n-1} + \cdots + a_2 \cdot 10^2 + a_1 \cdot 10 + a_0,$$

$$a_i \in \{0, 1, \dots, 9\}, a_n \neq 0.$$

- Divisibility by 3 and 9: $10^k - 1 = 99 \dots 9 \Rightarrow 3 \mid (10^k - 1), 9 \mid (10^k - 1)$

$$\begin{aligned} A &= a_n \cdot 10^n + a_{n-1} \cdot 10^{n-1} + \cdots + a_2 \cdot 10^2 + a_1 \cdot 10 + a_0 = \\ &= a_n(10^n - 1) + a_{n-1}(10^{n-1} - 1) + \cdots + a_1(10 - 1) + \\ &\quad + a_n + a_{n-1} + \cdots + a_1 + a_0 \end{aligned}$$

$\Rightarrow A$ is divisible by 3 or 9 if the sum of its digits is divisible by 3 or 9.

Divisibility rules

$A \in \mathbb{N} \Rightarrow$

$$A = a_n \cdot 10^n + a_{n-1} \cdot 10^{n-1} + \cdots + a_2 \cdot 10^2 + a_1 \cdot 10 + a_0,$$

$$a_i \in \{0, 1, \dots, 9\}, a_n \neq 0.$$

- Divisibility by 11: $10^1 + 1 = 11$, $10^2 - 1 = 99$, $10^3 + 1 = 1001$, $10^4 - 1 = 9999$, ... We can prove that

$11 \mid (10^k + 1)$ if k is odd and $11 \mid (10^k - 1)$ if k is even.

$$\begin{aligned} A &= a_0 + a_1(10^1 + 1) - a_1 + a_2(10^2 - 1) + a_2 + \cdots = \\ &= (a_1(10^1 + 1) + a_2(10^2 - 1) + \cdots) + (a_0 - a_1 + a_2 - a_3 + \cdots) \end{aligned}$$

$\Rightarrow A$ is divisible by 11 if the alternating sum of its digits is divisible by 11

Definition

We say that $d \in \mathbb{N}$ is the **greatest common divisor** of the integers a and b

- $d|a$ and $d|b$,
- for all $\bar{d} \in \mathbb{N}$ such that $\bar{d}|a$ and $\bar{d}|b$, the relation $\bar{d}|d$ also holds.

Notation: $d = \gcd(a, b)$.

Furthermore $d \in \mathbb{N}$ is the **greatest common divisor** of $a_1, a_2, \dots, a_n \in \mathbb{Z}$ if

- $d|a_i$, $i \in \{1, \dots, n\}$,
- for every $\bar{d} \in \mathbb{N}$ such that $\bar{d}|a_i$ ($i \in \{1, \dots, n\}$), the relation $\bar{d}|d$ also holds.

Definition

The integers a and b are called **relatively prime** or **coprime numbers** if $\gcd(a, b) = 1$.

Definition

We say that $k \in \mathbb{N}$ is the **least common multiple** of $a_1, a_2, \dots, a_n \in \mathbb{Z}$ if

- $a_i|k$, $i \in \{1, \dots, n\}$,
- for all $\bar{k} \in \mathbb{N}$ such that $a_i|\bar{k}$ ($i \in \{1, \dots, n\}$), the property $k|\bar{k}$ also holds.

Notation: $k = \text{lcm}(a_1, a_2, \dots, a_n)$.

The Euclidean algorithm

Theorem – Euclidean division

Given arbitrary $a, b \in \mathbb{Z}$, $b \neq 0$ numbers there uniquely exist integers $q, r \in \mathbb{Z}$ such that

$$a = b \cdot q + r, \quad 0 \leq r < |b|.$$

The Euclidean algorithm (or Euclid's algorithm)

$a, b \in \mathbb{Z}$, $b \neq 0$, theorem above $\Rightarrow q, r \in \mathbb{Z}$, let us denote them by q_0, r_0 this time:

$$a = b \cdot q_0 + r_0$$

Let us repeat the Euclidean division with b and $r_0 \Rightarrow q_1, r_1 \in \mathbb{Z}$, then with r_0 and $r_1 (\Rightarrow q_2, r_2 \in \mathbb{Z})$:

$$b = r_0 \cdot q_1 + r_1$$

$$r_0 = r_1 \cdot q_2 + r_2.$$

By continuing the procedure in this manner (each time with the obtained remainders) we finish in finite steps, since

$$|b| > r_0 > r_1 > r_2 > \cdots > r_i > \cdots \geq 0.$$

Theorem

When applying the Euclidean algorithm for the integers a and $b \neq 0$, the last non-zero remainder is the greatest common divisor of a and b . Furthermore, if $d := \gcd(a, b)$, then the equation

$$ax + by = d$$

can be solved among integers. That is, there exist $x, y \in \mathbb{Z}$ solutions.

Example: $\gcd(1227, 216) = ?$, $\gcd(-1227, -216) = ?$

$$1227 = 216 \cdot 5 + 147$$

$$216 = 147 \cdot 1 + 69$$

$$147 = 69 \cdot 2 + 9$$

$$69 = 9 \cdot 7 + 6$$

$$9 = 6 \cdot 1 + 3$$

$$6 = 3 \cdot 2 + 0$$

Thus, $\gcd(1227, 216) = 3$.

Linear Diophantine equations

Definition

Equations of the form $ax + by = c$ (where $a, b, c \in \mathbb{Z}$ are known, $x, y \in \mathbb{Z}$ are unknown) are called **linear Diophantine equations**.

Theorem

The linear Diophantine equation $ax + by = c$ is solvable if, and only if, $\gcd(a, b) \mid c$.

Theorem

If the Diophantine equation $ax + by = c$ is solvable, then it has infinitely many solutions, which can be written in the form

$$x = x_0 + t \frac{b}{\gcd(a, b)}, \quad y = y_0 - t \frac{a}{\gcd(a, b)}, \quad t \in \mathbb{Z}$$

where (x_0, y_0) is a particular solution.

Example: Solve the Diophantine equations below, if possible.

$$147x + 69y = 3$$

$$140x + 322y = 98$$

Prime numbers

Every $n > 1$, $n \in \mathbb{N}$ has two positive divisors: 1 and n , these are called the **trivial divisors** of n . All the other divisors are called **non-trivial divisors**.

Definition

Natural numbers which are greater than 1 and has only trivial divisors are called **prime numbers** or **primes**. Natural numbers with also non-trivial divisors are called **composite numbers**. 1 is a **unit**.

Theorem

An integer $p > 1$ is prime if, and only if, $p|ab$ implies $p|a$ or $p|b$.

Theorem – the fundamental theorem of arithmetic (also called unique-prime-factorization theorem)

Every natural number greater than 1 is either a prime itself or is the product of prime numbers. Furthermore, this product is unique up to the order of the factors. The obtained unique product is called the **canonical representation** or the **standard form** of n , which is $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_r^{\alpha_r}$, where $p_1, p_2, \dots, p_r$ are pairwise different primes, $\alpha_1, \alpha_2, \dots, \alpha_r \in \mathbb{N}$.

Example

Determine $\gcd(1260, 14850)$ and $\text{lcm}(1260, 14850)$.

$$1260 = 2^2 \cdot 3^2 \cdot 5 \cdot 7 \text{ and } 14850 = 2 \cdot 3^3 \cdot 5^2 \cdot 11$$

Greatest common divisor: take the common prime factors on the smaller power.

$$1260 = 2^2 \cdot 3^2 \cdot 5^1 \cdot 7 \text{ and } 14850 = 2^1 \cdot 3^3 \cdot 5^2 \cdot 11$$

$$\gcd(1260, 14850) = 2 \cdot 3^2 \cdot 5 = 90$$

Least common multiple: take all prime factors on the greater power.

$$1260 = 2^2 \cdot 3^2 \cdot 5 \cdot 7^1 \text{ and } 14850 = 2 \cdot 3^3 \cdot 5^2 \cdot 11^1$$

$$\text{lcm}(1260, 14850) = 2^2 \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 11 = 207900$$

Number of divisors

Theorem

The number of positive divisors of a natural number $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_r^{\alpha_r}$ is

$$d(n) = (\alpha_1 + 1)(\alpha_2 + 1) \dots (\alpha_r + 1).$$

Example: $1260 = 2^2 \cdot 3^2 \cdot 5 \cdot 7$ and $14850 = 2 \cdot 3^3 \cdot 5^2 \cdot 11$

Theorem

There are infinitely many prime numbers.

Proof: Suppose that there are only finitely many prime numbers, let them be $p_1, p_2, \dots, p_k$. Consider the number $b = p_1 \cdot p_2 \cdot \dots \cdot p_k + 1$. Then $b \neq 1$ and b is a composite number, thus for some index $i \in \{1, 2, \dots, k\}$ we have $p_i | b$. But $p_i | \prod p_j$ as well, thus $p_i | 1$, which is a contradiction.

Remark

The integers a and b are coprime numbers if there are no common prime factors in their canonical representation.

Congruence

Let $a, b \in \mathbb{Z}$, $m \in \mathbb{N}$.

Definition

We say that a and b are congruent modulo m if $m \mid (a - b)$.

Notation: $a \equiv b \pmod{m}$, m : is the modulus of the congruence.

Example: for $m = 4$ we have $3 \equiv 11 \pmod{4}$

The integers $a, b \in \mathbb{Z}$ are congruent modulo m if they provide the same remainder when divided by m .

Theorem

The congruence modulo m is a so-called equivalence relation: reflexive, symmetric, transitive.

Definition

Let us consider the class of integers which are congruent with each other modulo m . The obtained classes are called the congruence classes or residue classes modulo m . The residue classes are represented by the integers $0, 1, \dots, m - 1$. Thus, there are m residue classes modulo m .

The properties of congruence

Proposition – the properties of congruence

Let $m \in \mathbb{N}$ ($m \geq 2$) and $a, b, c, d \in \mathbb{Z}$.

① If $a \equiv b$ and $c \equiv d \pmod{m}$, then

$$a \pm c \equiv b \pm d \pmod{m} \quad \text{and} \quad a \cdot c \equiv b \cdot d \pmod{m}.$$

② If $a \cdot c \equiv b \cdot c \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b$.

Example: $15 \equiv 63 \pmod{8}$ and $10 \equiv 18 \pmod{8}$

Definition

Any set of m integers, no two of which are congruent modulo m , is called a **complete residue system** modulo m . The set of integers $\{0, 1, 2, \dots, m-1\}$ is called the **least residue system** modulo m .

Example: for $m = 5$ the set $\{5, 6, 12, 28, -6\}$ is a complete residue system, while $\{0, 1, 2, 3, 4\}$ is the least residue system.

Proposition

If $a \equiv b \pmod{m}$, then $\gcd(a, m) = \gcd(b, m)$.

Reduced residue system

Definition

A residue class is a member of the **reduced residue system** if its members are coprime to the modulus. Notation: the number of elements of a reduced residue system **modulo** m is denoted by $\varphi(m)$. That is

$$\varphi(m) = \#\{a \in \{1, \dots, m\} \mid \gcd(a, m) = 1\}.$$

The name of the function φ : **Euler's φ function** or **Euler's totient function**.

By definition, $\varphi(1) = 1$.

Examples: cardinality of the reduced residue system:

m	complete	reduced	$\varphi(m)$
$m = 2$	0,1	1	$\varphi(2) = 1$
$m = 3$	0,1,2	1,2	$\varphi(3) = 2$
$m = 4$	0,1,2,3	1,3	$\varphi(4) = 2$
$m = 5$	0,1,2,3,4	1,2,3,4	$\varphi(5) = 4$
$m = 6$	0,1,2,3,4,5	1,5	$\varphi(6) = 2$
$m = 7$	0,1,2,3,4,5,6	1,2,3,4,5,6	$\varphi(7) = 6$

Euler's φ function

Proposition

If p is a prime, then $\varphi(p) = p - 1$.

Theorem

The value of Euler's φ function can be calculated by the formula

$$\varphi(m) = m \cdot \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right),$$

where m has canonical representation $m = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_r^{\alpha_r}$.

Example: $m = 24$, $\varphi(24) = ?$

Theorem – Euler's theorem

If $\gcd(a, m) = 1$, then $a^{\varphi(m)} \equiv 1 \pmod{m}$.

Corollary – Fermat's little theorem

If p is a prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

Example: what is the remainder when dividing 2^{2025} by 15?

Congruence equations

Theorem

The (linear) congruence equation $ax \equiv b \pmod{m}$ is solvable among integers if, and only if, $\gcd(a, m) \mid b$.

Proof: we can derive a Diophantine equation from the congruence equation:

$$\begin{aligned} ax \equiv b \pmod{m} &\Leftrightarrow m \mid (ax - b) \Leftrightarrow \\ &\Leftrightarrow \exists y \in \mathbb{Z} : my = ax - b \Leftrightarrow ax - my = b \end{aligned}$$

Remark: if $c \in \mathbb{Z}$ is a solution, then so is $c + km$.

Example: $12x \equiv 8 \pmod{16}$

$\gcd(12, 16) = 4 \mid 8 \Rightarrow$ the equation is solvable

Example

Solve the linear congruence equation $12x \equiv 8 \pmod{16}$.

$\gcd(12, 16) \mid 8 \Rightarrow$ the equation is solvable

Solution 1:

Solve the linear Diophantine equation $12x - 16y = 8$ (i.e., $3x - 4y = 2$).

Solution 2:

Consider the equation $\frac{12}{\gcd(12,16)}x \equiv \frac{8}{\gcd(12,16)} \pmod{\frac{16}{\gcd(12,16)}}$.

Then

$$3x \equiv 2 \pmod{4}$$

$$3x \equiv 2 + 4 = 6 \pmod{4}$$

$$x \equiv 2 \pmod{4} \quad (\text{because } \gcd(3, 4) = 1)$$

The solutions:

$$x = \dots, -10, -6, -2, 2, 6, 10, \dots$$

Complex numbers

Looking for solutions of equations in different sets:

- $\mathbb{N}$: $5 + x = 3 \Rightarrow$ not solvable
- $\mathbb{Z}$: $5 \cdot x = 3 \Rightarrow$ not solvable
- $\mathbb{Q}$: $x^2 = 3 \Rightarrow$ not solvable
- $\mathbb{R}$: $x^2 = -1 \Rightarrow$ not solvable

$\leadsto$ Let's „extend“ $\mathbb{R}$ with $\sqrt{-1}$.

Notation, definition

The symbol $i := \sqrt{-1}$ is the **imaginary unit**.

Definition

Numbers of the form $a + bi$ where $a, b \in \mathbb{R}$ and $i^2 = -1$, are called **complex numbers**. The set of complex numbers is denoted by $\mathbb{C}$.

Let $z = a + bi \in \mathbb{C}$. This is called the **algebraic form** of z .

$a = \operatorname{Re}(z)$: **real part** of z

$b = \operatorname{Im}(z)$: **imaginary part** of z .

Operations with the algebraic form complex numbers

Operations in $\mathbb{C}$

If $z = a + bi$ and $w = c + di$ are complex numbers, then

$$z + w := (a + c) + (b + d)i,$$

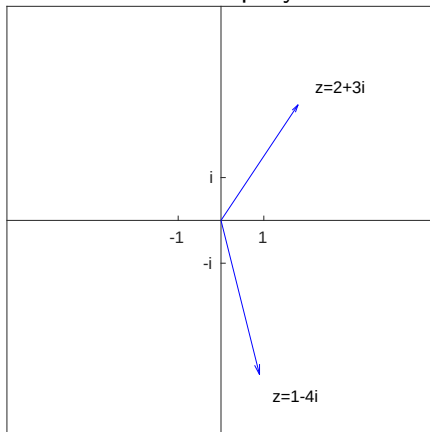
$$z \cdot w := (ac - bd) + (ad + bc)i.$$

Example:

- Let $z = 2 - 3i$ and $w = -1 + 4i$. Calculate $z + w$, and zw .
- Provide the algebraic form of i^{13} , $(2 - i)^2$ and $(1 + 2i)^3$.
- What is the algebraic form of $(5 - 2i)(5 + 2i)$?

Visual representation: on the complex plane

The complex number $z = a + bi$ is uniquely determined by the pair (a, b) .



Definition

The **absolute value** of the complex number $z = a + bi$ is $|z| := \sqrt{a^2 + b^2}$.

The **complex conjugate** of $z = a + bi$ is defined as $\bar{z} = a - bi$.

Then $z \cdot \bar{z} = |z|^2$.

The trigonometric form of complex numbers

A complex number $z = a + bi$ is uniquely determined by a and b , but by two other values as well:

- the **absolute value** of z : $r := |z| := \sqrt{a^2 + b^2}$
- the **argument** of z : φ , which is the angle of the vector (a, b) with the positive real axis. Then

$$a = r \cdot \cos \varphi,$$

$$b = r \cdot \sin \varphi,$$

and the **trigonometric form** of z is

$$z = r \cdot (\cos \varphi + i \cdot \sin \varphi),$$

where $\varphi \in [0, 2\pi[$.

Trigonometric form

Example

What is the algebraic form of the complex number which has absolute value 3 and argument $\frac{\pi}{4}$?

Example

What are the trigonometric forms of the complex numbers 2 , -5 , $3i$, $-2i$, $1 + \sqrt{3}i$, $-2 + 2i$, $-\sqrt{3} - i$ and $1 - i$?

Example

What are the trigonometric forms of 2 , -5 , $3i$, $-2i$?

1. If $z = 2$, then $|z| = 2$ and the complex number is on the positive part of the real axis, i.e., $\varphi = 0$. Hence,

$$z = 2(\cos 0 + i \sin 0)$$

2. If $z = -5$, then $|z| = 5$ and the complex number is on the negative part of the real axis, i.e., $\varphi = \pi$. Hence,

$$z = 5(\cos \pi + i \sin \pi)$$

3. If $z = 3i$, then $|z| = 3$ and the complex number is on the positive part of the imaginary axis, i.e., $\varphi = \frac{\pi}{2}$. Hence,

$$z = 3 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)$$

4. If $z = -2i$, then $|z| = 2$ and the complex number is on the negative part of the imaginary axis, i.e., $\varphi = \frac{3\pi}{2}$. Hence,

$$z = 2 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right)$$

Example

What are the trigonometric forms of $1 + \sqrt{3}i$, $-2 + 2i$, $-\sqrt{3} - i$ and $1 - i$?

Solution.

Step 1. Calculate the value of $|z|$ and decide which quadrant of the complex plane z lies in: plot it on the complex plane or use the table

real part (cos)	+	-	-	+
imaginary part (sin)	+	+	-	-
quadrant	1st	2nd	3rd	4th

Step 2. By using the formula $\cos \varphi_0 = \frac{|a|}{|z|}$ (or $\sin \varphi_0 = \frac{|b|}{|z|}$), determine the value of φ_0 , such that $0 < \varphi_0 < \frac{\pi}{2}$.

Step 3. The trigonometric form of z is $z = |z|(\cos \varphi + i \sin \varphi)$, where

- $\varphi = \varphi_0$, when z lies in the first quadrant,
- $\varphi = \pi - \varphi_0$, when z lies in the second quadrant,
- $\varphi = \pi + \varphi_0$, when z lies in the third quadrant,
- $\varphi = 2\pi - \varphi_0$, when z lies in the fourth quadrant.

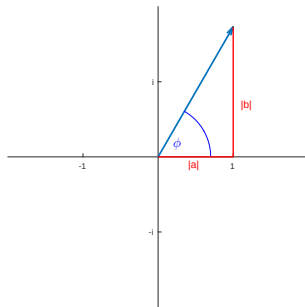
When $z = 1 + \sqrt{3}i$, then

Step 1. $|z| = 2$

Step 2. $\cos \varphi_0 = \frac{|a|}{|z|} = \frac{1}{2}$, so $\varphi_0 = \frac{\pi}{3}$

Step 3. z lies in the first quadrant, hence $\varphi = \varphi_0 = \frac{\pi}{3}$, i.e., the trigonometric form of z is

$$z = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$



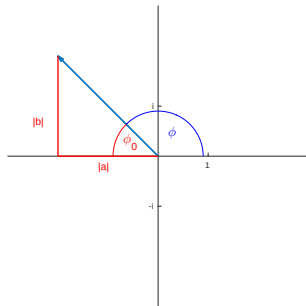
When $z = -2 + 2i$, then

Step 1. $|z| = \sqrt{8}$

Step 2. $\cos \varphi_0 = \frac{|a|}{|z|} = \frac{2}{\sqrt{8}} = \frac{1}{\sqrt{2}}$, hence $\varphi_0 = \frac{\pi}{4}$

Step 3. z lies in the second quadrant, hence $\varphi = \pi - \varphi_0 = \frac{3\pi}{4}$, i.e., the trigonometric form of z is

$$z = \sqrt{8} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$



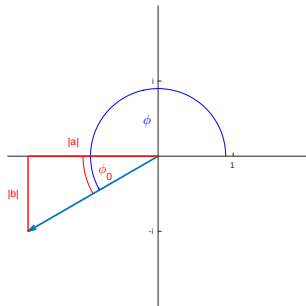
When $z = -\sqrt{3} - i$, then

Step 1. $|z| = 2$

Step 2. $\cos \varphi_0 = \frac{|a|}{|z|} = \frac{\sqrt{3}}{2}$, hence $\varphi_0 = \frac{\pi}{6}$

Step 3. z lies in the third quadrant, hence $\varphi = \pi + \varphi_0 = \frac{7\pi}{6}$, i.e., the trigonometric form of z is

$$z = 2 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right)$$



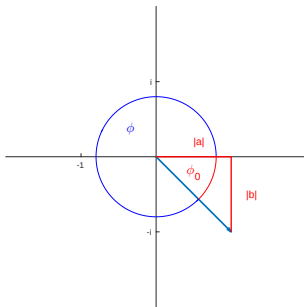
When $z = 1 - i$, then

Step 1. $|z| = \sqrt{2}$

Step 2. $\cos \varphi_0 = \frac{|a|}{|z|} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$, hence $\varphi_0 = \frac{\pi}{4}$

Step 3. z lies in the fourth quadrant, hence $\varphi = 2\pi - \varphi_0 = \frac{7\pi}{4}$, i.e., the trigonometric form of z is

$$z = \sqrt{2} \left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \right)$$



Operations with the trigonometric form

Let

$$z_1 = r_1 \cdot (\cos \varphi_1 + i \cdot \sin \varphi_1) \quad \text{and} \quad z_2 = r_2 \cdot (\cos \varphi_2 + i \cdot \sin \varphi_2).$$

- Multiplication: $z_1 \cdot z_2 = r_1 r_2 \cdot (\cos(\varphi_1 + \varphi_2) + i \cdot \sin(\varphi_1 + \varphi_2))$
- Division: $\frac{z_1}{z_2} = \frac{r_1}{r_2} \cdot (\cos(\varphi_1 - \varphi_2) + i \cdot \sin(\varphi_1 - \varphi_2))$
- Powers: if $z = r \cdot (\cos \varphi + i \cdot \sin \varphi)$ and $n \in \mathbb{Z}$, then
$$z^n = r^n \cdot (\cos(n\varphi) + i \cdot \sin(n\varphi)) \quad (\text{Moivre's formula}).$$

Example:

- Determine the values of $z_1 \cdot z_2$, $\frac{z_1}{z_2}$ and z_1^4 if

$$z_1 = 3\left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5}\right), \quad z_2 = 2\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$$

- $z = 1 - i$, $z^{60} = ?$

Determining the n^{th} roots in $\mathbb{C}$

$$\sqrt[n]{z} = ?$$

The equation $x^n = z$ (where x is the unknown parameter) has solutions as the n th roots of z , and there are n of them. $\rightsquigarrow w_0, w_1, \dots, w_{n-1}$

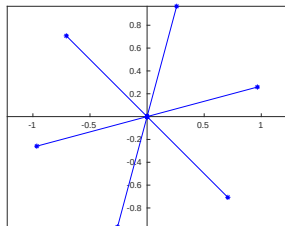
Theorem – n^{th} roots of a complex number

If $z = |z| \cdot (\cos \varphi + i \sin \varphi)$ and $n \in \mathbb{N}$, then the equation $x^n = z$ has exactly n solutions, these are the n^{th} roots of z . They can be calculated as

$$w_k = \sqrt[n]{|z|} \cdot \left(\cos \frac{\varphi + 2k\pi}{n} + i \cdot \sin \frac{\varphi + 2k\pi}{n} \right), \quad k = 0, 1, \dots, n-1.$$

The 6th roots of $z = i = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$ are

$$\begin{aligned} w_0 &= \cos \frac{\pi}{12} + i \sin \frac{\pi}{12}, & w_1 &= \cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12} \\ w_2 &= \cos \frac{9\pi}{12} + i \sin \frac{9\pi}{12}, & w_3 &= \cos \frac{13\pi}{12} + i \sin \frac{13\pi}{12} \\ w_4 &= \cos \frac{17\pi}{12} + i \sin \frac{17\pi}{12}, & w_5 &= \cos \frac{21\pi}{12} + i \sin \frac{21\pi}{12} \end{aligned}$$



Another example for calculating the n^{th} roots

Formula for n^{th} roots of a complex number

If $z = |z| \cdot (\cos \varphi + i \sin \varphi)$ and $n \in \mathbb{N}$, then

$$w_k = \sqrt[n]{|z|} \cdot \left(\cos \frac{\varphi + 2k\pi}{n} + i \sin \frac{\varphi + 2k\pi}{n} \right), \quad k = 0, 1, \dots, n-1.$$

Example. $z = -\sqrt{3} - i$, $\sqrt[3]{z} = ?$

We have calculated (on Slide 47) the trigonometric form of z :

$$z = 2 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right).$$

Thus here $|z| = 2$, $\varphi = \frac{7\pi}{6}$ and $n = 3 \Rightarrow k = 0, 1, 2$.

So the 3rd roots of z are

$$w_0 = \sqrt[3]{2} \left(\cos \frac{7\pi}{18} + i \sin \frac{7\pi}{18} \right),$$

$$w_1 = \sqrt[3]{2} \left(\cos \frac{19\pi}{18} + i \sin \frac{19\pi}{18} \right), \quad w_2 = \sqrt[3]{2} \left(\cos \frac{31\pi}{18} + i \sin \frac{31\pi}{18} \right).$$

Roots of unity

Definition

The n^{th} roots of the (complex) number 1 are called the n^{th} roots of unity. Thus, these are the solutions of the equation $x^n = 1$.

Since

$$z = 1 = 1 + 0 \cdot i = \cos 0 + i \cdot \sin 0,$$

and the n^{th} roots of a complex number are given by the formula

$$\sqrt[n]{r} \cdot \left(\cos \frac{\varphi + 2k\pi}{n} + i \cdot \sin \frac{\varphi + 2k\pi}{n} \right), \quad k = 0, 1, \dots, n-1,$$

the n^{th} roots of unity are

$$\varepsilon_k = \cos \frac{2k\pi}{n} + i \cdot \sin \frac{2k\pi}{n}, \quad k = 0, 1, \dots, n-1.$$

Remark: $\forall n \in \mathbb{N}$ we have $\varepsilon_0 = 1$.

Examples: what are the n th roots of unity in the cases $n = 2$, $n = 3$ and $n = 4$? Plot them on the complex plane.

Solving second order equations in $\mathbb{C}$

Second order polynomials with real coefficients

$ax^2 + bx + c = 0$, $a \neq 0$, $a, b, c \in \mathbb{R}$. Possible cases:

- 1 $b^2 - 4ac > 0 \Rightarrow$ two real solutions
- 2 $b^2 - 4ac = 0 \Rightarrow$ one real solution with multiplicity 2
- 3 $b^2 - 4ac < 0 \Rightarrow$ two complex solutions, which are complex conjugates of each other:

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-b}{2a} \pm i \cdot \frac{\sqrt{4ac - b^2}}{2a}$$

Example. Solve the equation $x^2 + 2x + 2 = 0$.

$$x_{1,2} = \frac{-2 \pm \sqrt{4 - 8}}{2} = \frac{-2 \pm \sqrt{-4}}{2} = \frac{-2 \pm 2\sqrt{-1}}{2} = -1 \pm i$$

Polynomials

Definition

Let x be a so-called **indeterminate** (or variable, a symbol). The expression

$$p(x) = \sum_{i=0}^n a_i x^i = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n,$$

where $a_i \in \mathbb{R}$, is called a **polynomial**.

- The set of polynomials with real coefficients is denoted by $\mathbb{R}[x]$.
- If $a_n \neq 0$, then n is the **degree** or **order** of the polynomial. Notation: $\deg(p) = n$.
- The real numbers a_i are called the **coefficients** of the polynomial, a_n is the **leading coefficient**.
- If $p(x) = a_0$, then it is a **zero-order** or **constant** polynomial.

Examples:

$$p_1(x) = 3 + 2x + x^4 + 3x^5 \rightarrow \text{degree 5 polynomial}$$

$$p_2(x) = 2 + x^3 + 3x^4 + 0 \cdot x^5 \rightarrow \text{degree 4 polynomial}$$

Operations with polynomials

Definition

Let us consider the polynomials

$$p(x) = a_0 + a_1x + \cdots + a_nx^n \quad \text{and} \quad q(x) = b_0 + b_1x + \cdots + b_mx^m.$$

- The two polynomials are equal if $n = m$ and $a_i = b_i$, $i = 0, 1, \dots, n$.
- The sum of the two polynomials if, e.g., $n > m$, is

$$(p + q)(x) := p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_m + b_m)x^m + a_{m+1}x^{m+1} + \cdots + a_nx^n.$$

- The product of the two polynomials is

$$(p \cdot q)(x) := p(x) \cdot q(x) = (a_0 \cdot b_0) + (a_0b_1 + a_1b_0)x + (a_0b_2 + a_1b_1 + a_2b_0)x^2 + \cdots + a_nb_mx^{m+n}.$$

Thus:

$$\deg(p + q) \leq \max\{\deg(p), \deg(q)\}$$

$$\deg(p \cdot q) = \deg(p) + \deg(q)$$

Operations with polynomials – examples

Let

$$p(x) = 3x^2 - x + 10 \quad \text{and} \quad q(x) = -3x^2 + 8x.$$

Then

$$(p + q)(x) := p(x) + q(x) = 7x + 10,$$

$$\begin{aligned}(p \cdot q)(x) &:= p(x) \cdot q(x) = -9x^4 + 24x^3 + 3x^3 - 8x^2 - 30x^2 + 80x \\ &= -9x^4 + 27x^3 - 38x^2 + 80x.\end{aligned}$$

Thus here

$$\deg(p + q) = 1 \leq 2 = \max\{\deg(p), \deg(q)\}$$

$$\deg(p \cdot q) = 4 = \deg(p) + \deg(q)$$

Euclidean division for polynomials

Theorem – Euclidean division for polynomials

If

$$p(x) = a_n x^n + \cdots + a_1 x + a_0,$$

$$s(x) = b_m x^m + \cdots + b_1 x + b_0,$$

where $a_n \neq 0$, $b_m \neq 0$ and $m < n$, then there exist uniquely polynomials $q(x)$ and $r(x)$ such that

$$p(x) = s(x) \cdot q(x) + r(x), \quad \deg(q) = n - m, \quad \deg(r) < m = \deg(s).$$

Example: $p(x) = x^4 + 3x^2 - 4$, $s(x) = x^2 + 2x$

Definition

If in the previous theorem $p(x) = s(x) \cdot q(x)$, that is, $r(x) = 0$, then $s(x)$ is a divisor of $p(x)$, which we denote by $s(x) | p(x)$.

Example: $p(x) = x^5 - 3x^4 + 4x + 1$, $s(x) = x^2 + x + 1$

The roots of polynomials

Definition

Let $p(x)$ be a polynomial with real (or complex) coefficients, that is, $p(x) \in \mathbb{R}[x]$ (or $p(x) \in \mathbb{C}[x]$). The number $x_0 \in \mathbb{C}$ is a **root** or **solution** of $p(x)$ if $p(x_0) = 0$.

Remark: If x_0 is a root of $p(x)$, then $(x - x_0) | p(x)$.

- $p(x) = x^3 + x^2 - 2x - 8$, since $p(2) = 0$, $x_0 = 2$ is a root
 $\implies x - 2$ is a divisor of $x^3 + x^2 - 2x - 8$
- For second-order polynomials:

$$ax^2 + bx + c = a(x - x_1)(x - x_2)$$

Definition

The **multiplicity** of the root x_0 in the polynomial $p(x)$ is k if $(x - x_0)^k | p(x)$, but $(x - x_0)^{k+1} \nmid p(x)$.

If $k = 1$, then x_0 is a **simple root**, if $k > 1$, it is a **multiple root** of $p(x)$.

E.g: What is the multiplicity of $x_0 = 1$ in the polynomial below?

$$p(x) = x^4 - 4x^3 + 2x^2 + 4x - 3$$

The Fundamental Thm of Algebra and its consequences

Theorem – the Fundamental Theorem of Algebra

Let $p(x) \in \mathbb{C}[x]$ be a non-constant polynomial. Then $\exists x_0 \in \mathbb{C} : p(x_0) = 0$, that is, $p(x)$ has a complex root.

Remark: Let $p(x) \in \mathbb{C}[x]$ be a polynomial of degree n ($n \geq 1$) and let $x_0 \in \mathbb{C}$ be a root of it. Then $(x - x_0) \mid p(x)$, thus:

$$p(x) = (x - x_0) \cdot q(x), \quad \text{where } \deg(q) = n - 1.$$

But the Fundamental Thm of Algebra holds also for $q(x)$, so there exists a root $x_1 \in \mathbb{C}$ of it, which implies the form $q(x) = (x - x_1) \cdot q_1(x)$, and that

$$p(x) = (x - x_0) \cdot (x - x_1) \cdot q_1(x); \quad \dots$$

Corollaries

- A degree n polynomial with complex coefficients has, counted with multiplicity, exactly n complex roots.
- A degree n polynomial with **real** coefficients has exactly n complex and **at most** n real roots.

The factored form of polynomials (over $\mathbb{R}$)

$$p(x) = a_n(x - x_1)^{\alpha_1} \dots (x - x_k)^{\alpha_k} \cdot (x^2 + b_1x + c_1)^{\beta_1} \dots (x^2 + b_lx + c_l)^{\beta_l}$$

where $(x - x_i)^{\alpha_i}$: linear (or first-order) factors

and $(x^2 + b_jx + c_j)^{\beta_j}$: second-order factors,

they cannot be factorized over $\mathbb{R}$

non-real (complex) roots are here,

which are pairwise conjugate

$$n = \deg(p) = \alpha_1 + \dots + \alpha_k + 2\beta_1 + \dots + 2\beta_l$$

If n is odd, then $\exists \alpha_i \neq 0$, so in this case $x_i \in \mathbb{R}$ is a root.

One more corollary of the Fundamental Thm of Algebra

- If $p(x)$ is an odd degree polynomial with real coefficients, then it has a real root.

The factored form over $\mathbb{C}$:

$$p(x) = a_n(x - x_1)^{\alpha_1} \dots (x - x_j)^{\alpha_j}, \quad \text{where } n = \alpha_1 + \dots + \alpha_j.$$

Importance of polynomials in computer science

- **Computational complexity:** Polynomial functions are used to describe the running time or computational complexity of algorithms. Polynomial Time: Algorithms with a complexity bounded by a polynomial are generally considered efficient.
- **Cryptography:** Many public-key cryptographic systems, like those based on elliptic curves rely on advanced concepts involving polynomial arithmetic.
- **Computer Graphics and Geometry:** In computer graphics, polynomials are used for modeling shapes and curves, e.g., Bézier curves and Splines.
- **Numerical Methods and Scientific Computing:** Many tasks involve approximating continuous functions with simpler forms. E.g., Lagrange or Newton polynomials, Taylor series.

Horner's method (or Horner's scheme)

Horner's method is an efficient algorithm for the evaluation of a polynomial. Efficient = fewer number of arithmetic operations.

Example

$$p(x) = 2x^5 + 3x^4 - 3x^2 + 5x - 1, \quad p(-2) = ?$$

Here $p(x)$ can be written as

$$p(x) = (((((2x + 3)x + 0)x - 3)x + 5)x - 1).$$

By substituting -2 , we get the following values in the different brackets:

	2	3	0	-3	5	-1
-2	2	-1	2	-7	19	-39

Thus $p(-2) = -39$.

Horner's method in general

So if $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$, then

$$p(x) = (\dots ((a_n x + a_{n-1}) \cdot x + a_{n-2}) \cdot x + \dots + a_1) \cdot x + a_0.$$

The algorithm:

$$b_n = a_n$$

$$b_{n-1} = b_n x^* + a_{n-1}$$

$$b_{n-2} = b_{n-1} x^* + a_{n-2}$$

...

$$b_0 = b_1 x^* + a_0 = p(x^*)$$

In tabular form

	a_n	a_{n-1}	a_{n-2}	$\dots$	a_2	a_1	a_0
x^*	b_n	b_{n-1}	b_{n-2}	$\dots$	b_2	b_1	$b_0 = p(x^*)$

- Number of arithmetic operations to perform **without** Horner's method: $n - 1 + n = 2n - 1$ multiplications and n additions.
- Number of arithmetic operations to perform **with** Horner's method: n multiplications, n additions.

Examples

- $p(x) = -x^5 - 3x^4 + 2x^3 - 5x + 2$, $p(2) = ?$
- Determine the quotient and the remainder when

$$p(x) = 2x^6 - 3x^4 - 5x^3 + x^2 - 4x - 4$$

is divided by $(x + 1)$.

When we divide by a **linear term**, that is, by $x - a$, the remainder will be a constant. We can apply Horner's scheme for a simpler notation:

	2	0	-3	-5	1	-4	-4
a	coefficients of quotient						remainder
-1	2	-2	-1	-4	5	-9	5

- Determine the multiplicity of $x_0 = 1$ in the polynomial

$$p(x) = 2x^5 - 4x^4 + 6x^3 - 14x^2 + 16x - 6.$$

Polynomials with integer coefficients

Theorem

The **integer roots** of a polynomial with integer coefficients divide the constant term of the polynomial.

Theorem

In the case of $\frac{p}{q}$ **rational roots** of a polynomial with integer coefficients p divides the constant term of the polynomial, while q divides the leading coefficient. (Here $\gcd(p, q) = 1$.)

Example: The roots of

$$p(x) = 30x^3 - 53x^2 + 31x - 6$$

are

$$\frac{1}{2}, \quad \frac{2}{3}, \quad \frac{3}{5}.$$

Rational functions

Definition

Let $p(x)$ and $q(x)$ be polynomials with real coefficients, and assume that they do not have a common root, that is, $\nexists x_0 \in \mathbb{R} : p(x_0) = q(x_0) = 0$. Assume, furthermore, that $q(x) \not\equiv 0$. The function $R(x) = \frac{p(x)}{q(x)}$ is called **rational function**.

Example: $\frac{2x+14}{x^4-6x^3+10x^2-6x+9} = \frac{2x+14}{(x-3)^2(x^2+1)}$

Theorem – Partial fraction decomposition

Let $R(x) = \frac{p(x)}{q(x)}$ be a rational function, where $q(x)$ has the factored form:

$$q(x) = b_m \cdot (x - x_1)^{\alpha_1} \dots (x - x_k)^{\alpha_k} \dots (x^2 + b_l x + c_l)^{\beta_l}.$$

There exist real numbers A_{ij}, B_{ij}, C_{ij} and a polynomial $s(x)$ such that

$$R(x) = s(x) + \sum_{i=1}^k \sum_{j=1}^{\alpha_i} \frac{A_{ij}}{(x - x_i)^j} + \sum_{i=1}^l \sum_{j=1}^{\beta_i} \frac{B_{ij}x + C_{ij}}{(x^2 + b_i x + c_i)^j}$$

Rational functions – Example

So the fraction

$$\frac{2x + 14}{x^4 - 6x^3 + 10x^2 - 6x + 9} = \frac{2x + 14}{(x - 3)^2(x^2 + 1)}$$

can be written as

$$\frac{2x + 14}{(x - 3)^2(x^2 + 1)} = \frac{A_1}{x - 3} + \frac{A_2}{(x - 3)^2} + \frac{Bx + C}{x^2 + 1}.$$

By bringing the terms to a common denominator, we obtain

$$2x + 14 = A_1(x - 3)(x^2 + 1) + A_2(x^2 + 1) + (Bx + C)(x - 3)^2,$$

from which we get $A_1 = -1$, $A_2 = 2$, $B = C = 1$. Thus

$$\frac{2x + 14}{(x - 3)^2(x^2 + 1)} = -\frac{1}{x - 3} + \frac{2}{(x - 3)^2} + \frac{x + 1}{x^2 + 1}.$$

Combinatorics – Permutation of distinct elements

Definition

Let A be a set with n distinct elements. By a **permutation** of A we mean an arrangement of all the members of A into some sequence.

Theorem

The number of all permutations of a set of n distinct elements is

$$P_n = n! = n(n-1)(n-2) \dots 2 \cdot 1.$$

Other notation: $P(n, n) = n!$.

Examples:

- (1) There are 10 participants at a running competition. How many different orders can they finish in, if there are no ties?
 - (2) How many 5 digit numbers can be formed from the digits 3,4,5,7,9, if each digit can be listed only once?
- What if we consider the digits 2,2,2,7,7?

Permutation of multisets or ordering with identical items

How many 5 digit numbers can be formed from the digits 2,2,2,7,7?

Solution: If we treat all the 2's and 7's as different numbers, then we had $5!$ possible orders. But by changing the orders of the 2's, for example, we get the same 5-digit number. $\Rightarrow$ We have to divide $5!$ by the possible orders of identical elements, so the result is:

$$\frac{5!}{2! \cdot 3!} = 10.$$

Theorem

If we consider n elements of k type, ℓ_1 from the first type, ℓ_2 from the second type, etc. (so $\ell_1 + \ell_2 + \dots + \ell_k = n$), then the number of all permutations of these n elements is

$$P_n^{\ell_1, \dots, \ell_k} = \frac{n!}{\ell_1! \dots \ell_k!}$$

Example: We have 2 red, 1 orange and 3 yellow flowers, which we want to put in our window. How many possibilities do we have for the order of the flowers?

Partial permutations or k -permutations of n

Definition and theorem

In the case of a k -permutation of n or partial permutation we consider arrangements of a fixed length k of elements taken from a given set of size n . Here each element can occur at most once. The number of these arrangements is

$$P(n, k) = nPk = \frac{n!}{(n-k)!} = n \cdot (n-1) \dots (n-k+1).$$

Here necessarily $n \geq k$.

So we choose k elements out of n and arrange them into some order
 $\rightsquigarrow$ ordered selections without repetition

Examples: (1) There are 10 participants at a running competition. How many possibilities are there for the podium (that is, for the first 3 places)?
(2) There is a game, where there are 5 different prizes and they choose the winners from 200 participants randomly. How many possibilities are there for choosing the winners if everyone can win at most 1 prize?
What if the participants can be chosen more than once?

Permutations with repetition

Definition and theorem

By **permutations with repetition** we mean ordered selections of the elements of a set of size n of length k where repetition is allowed. These are also called **k -tuples**. The number of all arrangements of this type is

$$P(n, k)^{\text{rep}} = n^k.$$

So we choose k elements and arrange them into some order, the elements can occur more than once. Thus here $n < k$ is possible as well.

↪ ordered selections with repetition

Examples: (1) In how many ways can one fill a toto coupon? (14 matches, 3 possible results: 1, 2, or X)

(2) How many subsets does a set with n elements have?

The subsets are in a one-to-one correspondence with binary sequences of length n : 100101...110.

$$P(2, n)^{\text{rep}} = 2^n \text{ possibilities.}$$

Combination without repetition

Definition and theorem

A **combination** is a way of selecting items from a collection, such that (unlike permutations) the order of selection does not matter. So we choose a subset of k elements of a set with n elements. The number of all ways to select k items out of n elements without regard to order of selection is:

$$C(n, k) = nCk = \frac{n!}{k!(n-k)!} =: \binom{n}{k}.$$

By definition $0! = 1$.

Here necessarily $n \geq k$.

Examples:

(1) Find the number of possible fillings of a lottery coupon (5 numbers from 90).

(2) There is a game, where there are 5 alike prizes and they choose the winners from 200 participants randomly. How many possibilities are there for choosing the winners if everyone can win at most 1 prize?

What if the participants can be chosen more than once?

Combination with repetition

Definition and theorem

A **k -combination with repetitions**, or **k -multicombination**, or **multisubset of size k** from a set S is given by a sequence of k not necessarily distinct elements of S , where order is not taken into account. The number of all ways to select k items out of n elements (with repetition) without regard to order of selection is:

$$C(n, k)^{\text{rep}} = \left(\binom{n}{k} \right) = \binom{n + k - 1}{k}.$$

Here $n < k$ is possible as well.

Examples: (1) In how many ways can we distribute 10 (similar) apples among 4 children?

(2) If we roll 3 (alike) dice, how many possible ways are there for the result (that is, for the distribution of the thrown numbers)?

Binomial theorem or binomial expansion

Proposition

Let $k, n \in \mathbb{N} \cup \{0\}$, $n \geq k$. Then

$$\binom{n}{k} = \binom{n}{n-k}.$$

Theorem – binomial theorem

Let $x, y \in \mathbb{C}$, $n \in \mathbb{N}$. Then

$$\begin{aligned}(x+y)^n &= \sum_{k=0}^n \binom{n}{k} x^k y^{n-k} = \\ &= \binom{n}{n} x^n + \binom{n}{n-1} x^{n-1} y + \binom{n}{n-2} x^{n-2} y^2 + \dots + \binom{n}{1} x y^{n-1} + \binom{n}{0} y^n.\end{aligned}$$

Definition

The expression $\binom{n}{k}$ is called a **binomial coefficient**.

Binomial coefficients

Properties

For all $n \in \mathbb{N}$ we have

$$\binom{n}{0} = \binom{n}{n} = 1, \quad \binom{n}{1} = n.$$

Proposition

For all $n \in \mathbb{N}$, $0 < k < n$ we have

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

$\rightsquigarrow$ Pascal's triangle

Inclusion-exclusion principle

For two arbitrary sets A and B we have

$$\#(A \cup B) = \#A + \#B - \#(A \cap B).$$

For three arbitrary sets A , B and C we have

$$\begin{aligned}\#(A \cup B \cup C) &= \#A + \#B + \#C \\ &\quad - (\#(A \cap B) + \#(B \cap C) + \#(C \cap A)) \\ &\quad + \#(A \cap B \cap C)\end{aligned}$$

Example: In a class of 32, the kids study the following foreign languages:

- 16 kids study English, 13 study German, 13 study French;
- 5 kids study both German and French, 7 German and English, 6 of them study English and French;
- 4 kids study all 3 languages.

Out of the class, how many children don't study any language?

Solution: Because of the formula above,

$$\#(E \cup G \cup F) = 16 + 13 + 13 - (5 + 7 + 6) + 4 = 28.$$

So $32 - 28 = 4$ children don't study any language.

Inclusion-exclusion principle in general

Inclusion-exclusion principle

Let A be a finite set, $A_1, A_2, \dots, A_k$ are subsets of A . Assume that we know the cardinalities of the sets $A_1, A_2, \dots$, and the cardinalities of their intersections. Let S_j denote the number of elements of A such that it is an element of at least j sets of $A_1, \dots, A_k$, $j \in \{1, 2, \dots, k\}$. That is:

- $S_1 = \#A_1 + \#A_2 + \dots + \#A_k$
- $S_2 = \#(A_1 \cap A_2) + \#(A_1 \cap A_3) + \dots = \sum_{1 \leq i < j \leq k} \#(A_i \cap A_j)$
- ...
- $S_k = \#(A_1 \cap A_2 \cap \dots \cap A_k)$

Furthermore, let $S_0 = \#(A \setminus (A_1 \cup \dots \cup A_k))$ and $S = \#A$. Then

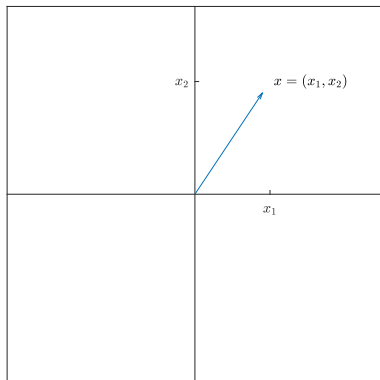
$$S = S_0 + S_1 - S_2 + S_3 - \dots + (-1)^{k-1} S_k.$$

In the previous exercise we had to calculate S_0 , the number of students not being in any of the set A_i (learning the i th language).

Linear algebra

Vectors

Vectors in the plane ($\mathbb{R}^2$)



Sum of two vectors:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \end{pmatrix}$$

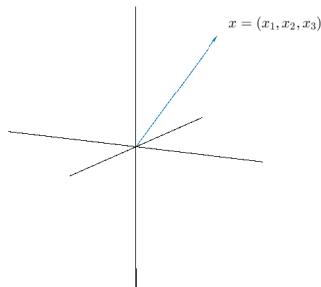
Product of a vector and a scalar:

$$\lambda \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \end{pmatrix} \quad (\lambda \in \mathbb{R})$$

The zero vector: $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Vectors

Vectors in space ($\mathbb{R}^3$)



Sum of two vectors:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ x_3 + y_3 \end{pmatrix}$$

Product of a vector and a scalar:

$$\lambda \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \\ \lambda x_3 \end{pmatrix} \quad (\lambda \in \mathbb{R})$$

The zero vector: $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

Vectors

Vectors in the n dimensional real space ($\mathbb{R}^n$)

Sum of two vectors:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

where $x_i \in \mathbb{R}$

The zero vector: $\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$

Product of a vector and a scalar:

$$\lambda \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \\ \vdots \\ \lambda x_n \end{pmatrix} \quad (\lambda \in \mathbb{R})$$

- Any two vectors can be added (the result is in $\mathbb{R}^n$).
- Any vector can be multiplied by an arbitrary real number, i.e., by a scalar (the result is in $\mathbb{R}^n$).

Thus $\mathbb{R}^n$ is closed under vector addition and scalar multiplication.

For the vector addition:

- commutativity: $x + y = y + x$
- associativity: $x + (y + z) = (x + y) + z$
- there exists a zero vector (0), such that: $x + 0 = x$
- for every vector there exists an additive inverse ($-x$): $x + (-x) = 0$

For the scalar multiplication:

- $(\lambda + \mu)x = \lambda x + \mu x$
- $\lambda(x + y) = \lambda x + \lambda y$
- $\lambda(\mu x) = (\lambda\mu)x$
- $1x = x$

Vector spaces

For every vector space it is true, that

- any two vectors can be added,
- any vector can be multiplied by a scalar,

and for the vector addition and scalar multiplication the previous 4-4 properties are true.

A vector space can consist not only of vectors in the classical sense, e.g., the set of real polynomials of degree at most n is a vector space.

$P_n[x]$: polynomials of degree at most n (with real coefficients)

When the scalars are from $\mathbb{R}$ then we call the vector space a **vector space over $\mathbb{R}$** .

The coordinates of a vector can be even complex numbers:

$$x \in \mathbb{C}^n, \quad \text{if} \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \text{where } x_i \in \mathbb{C}$$

The vector addition and scalar multiplication can be defined in the same way as in the case of real vectors (here the scalar can be a complex number, as well), the zero vector is the same: all coordinates are 0.

$\mathbb{C}^n$ is a vector space over $\mathbb{C}$

Subspace

Definition

A non-empty subset W of the vector space V is called a **linear subspace** (or simply a **subspace**) of V if it is a vector space itself, that is, W is closed under vector addition and scalar multiplication.

Example:

$$W_1 = \{x \in \mathbb{R}^3 : x_3 = 0\}$$

is a linear subspace of $\mathbb{R}^3$. W_1 consists of vectors having the form

$$\begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix}$$

- the sum of two vectors of this form has this form, too.
- a vector of this form multiplied by a scalar is also a vector of this form.

Example:

$$W_2 = \{x \in \mathbb{R}^3 : x_1 = x_2 = x_3\}$$

is a subspace of $\mathbb{R}^3$. W_2 consists of vectors with identical coordinates.

- The sum of two vectors of this form has this form, too.
- A vector of this form multiplied by a scalar is also a vector of this form.

Example:

$$W_3 = \{x \in \mathbb{R}^3 : x_3 = 1\}$$

is **not** a subspace of $\mathbb{R}^3$. W_3 consists of vectors of the form

$$\begin{pmatrix} x_1 \\ x_2 \\ 1 \end{pmatrix}.$$

- The sum of two vectors of this form **does not** have this form.

Remark

- $\{0\}$ and V are linear subspaces of V , called **trivial subspaces**.
- The zero vector is an element of every subspace.

Linear combination

The vector space is closed under vector addition and scalar multiplication, so for any $v_1, v_2, \dots, v_k$ vectors from the vector space their arbitrary weighted sum

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_k v_k$$

belongs to the vector space as well.

Definition

Let $v_1, v_2, \dots, v_k$ be vectors in V . The **linear combinations** of them are vectors of the form

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_k,$$

where $\lambda_1, \lambda_2, \dots, \lambda_k$ are scalars.

Trivial linear combination:

$$0 \cdot v_1 + \dots + 0 \cdot v_k = 0.$$

Examples

1. In $\mathbb{R}^2$, a linear combination of $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is

$$2v_1 + 3v_2 = 2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Prove, that every vector from $\mathbb{R}^2$ can be obtained as a linear combination of v_1 and v_2 .

2. In $\mathbb{R}^2$, a linear combination of $v_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$:

$$2v_1 - v_2 = 2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

Which vectors in $\mathbb{R}^2$ can be obtained as a linear combination of v_1 and v_2 ?

Examples

3. Which vectors in $\mathbb{R}^3$ can be obtained as a linear combination of v_1 and v_2 ?

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

4. Which vectors in $\mathbb{R}^3$ can be obtained as a linear combination of v_1 and v_2 ?

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} -2 \\ -2 \\ -2 \end{pmatrix}$$

Generated subspace, generating set

Theorem and definition

Let $v_1, v_2, \dots, v_k$ be vectors in V . Then the set of all linear combinations of these vectors form a linear subspace of V , called the **subspace spanned** or **generated** by $v_1, v_2, \dots, v_k$. Notation: $\mathcal{L}(v_1, \dots, v_k)$.

Definition

Let $\mathcal{G}$ be a set of vectors of V . We say that $\mathcal{G}$ is a **generating set** or that it **generates** V if the spanned subspace of $\mathcal{G}$ is the whole vector space.

Then every vector of V can be obtained as a linear combination of $\mathcal{G}$.

Example: Let $V = \mathbb{R}^2$, and consider the vectors

$$v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad w = \begin{pmatrix} 0 \\ 3 \end{pmatrix} \quad \text{and} \quad u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Here $\{v, w\}$ generates $\mathbb{R}^2$. Consequently, $\{u, v, w\}$ generates $\mathbb{R}^2$ as well, but considering the latter generating set, some vectors of $\mathbb{R}^2$ can be expressed as a linear combination of $\{u, v, w\}$ in more than one ways, e.g.,

$$\begin{pmatrix} 2 \\ 4 \end{pmatrix} = v + w = 2u + \frac{4}{3}w.$$

Linearly dependent and independent vectors

Definition

Let $v_1, v_2, \dots, v_k$ be vectors in V . We say that these vectors are **linearly dependent** if there exist scalars $\lambda_1, \lambda_2, \dots, \lambda_k \in \mathbb{R}$ **not all 0**, such that

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_k v_k = 0.$$

(Thus if the zero vector can be obtained as a non-trivial linear combination of the vectors.)

Otherwise we say that the vectors are **linearly independent**.

Remark: So in case of linear independence the condition

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_k v_k = 0$$

implies that $\lambda_i = 0, \forall i \in \{1, \dots, k\}$.

Example:

- Two non-zero vectors are linearly dependent if, and only if, they are scalar multiples of each other.
- Three vectors in $\mathbb{R}^3$ are linearly dependent if they lie in a common plane (through the origin).

Properties of linear dependence

Theorem

A set of vectors is linearly dependent if, and only if, some of the vectors can be obtained as a linear combination of the rest of the vectors.

Proposition

Consider a fixed set of vectors in a vector space.

- 1 If two (or more) of the vectors are the same, then the vectors are linearly dependent.
- 2 If one of the vectors is a scalar multiple of another vector, then the vectors are linearly dependent.
- 3 If the zero vector is among the vectors, then the vectors are linearly dependent. That is, a linearly independent set of vectors cannot contain the zero vector.
- 4 If a subset of the vectors is linearly dependent, then the entire set is linearly dependent.

Basis

Definition

A **basis** of V is a linearly independent set of vectors which generates V .

Basis: minimal set of vectors spanning the whole vector space.

- If $\mathcal{B}$ is a basis, then all elements of V can be **uniquely expressed as a linear combination** of the elements of $\mathcal{B}$.
- There are infinitely many bases of V .

Example. Some bases in $\mathbb{R}^2$:

$$\mathcal{B}_1 = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}, \quad \mathcal{B}_2 = \left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 5 \end{pmatrix} \right\}, \quad \mathcal{B}_3 = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}$$

Remark. The basis $\mathcal{B}_1$ is called the **natural basis** of $\mathbb{R}^2$.

The **natural basis of $\mathbb{R}^n$** can be defined in a similar way: all the coordinates of the i th vector are 0, but the i th, which is 1. ($i = 1, \dots, n$).

Theorem

If we find n linearly independent vectors in $\mathbb{R}^n$, then it is a basis.

Dimension and coordinates

Theorem and definition

Given a vector space V , all of its bases have the same cardinality (consist of the same number of vectors). This number is called the **dimension** of the vector space. Notation: $\dim(V)$.

Remark: if $V = \{0\}$, then $\dim(V) = 0$.

Definition

Let V be a vector space, $\mathcal{B} = \{b_1, \dots, b_n\}$ is one of its bases. Then all $v \in V$ can be uniquely expressed as a linear combination of the elements of $\mathcal{B}$, thus there exist unique scalars $\lambda_1, \lambda_2, \dots, \lambda_n$, such that

$$v = \lambda_1 b_1 + \dots + \lambda_n b_n.$$

These scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ are called the **coordinates** of v with respect to the basis $\mathcal{B}$.

Rank

Definition

Let $\mathcal{A}$ be a set of vectors. The **rank** of $\mathcal{A}$ is the dimension of the generated vector space:

$$\text{rank}(\mathcal{A}) = \dim(\mathcal{L}(\mathcal{A})).$$

Example. Determine the rank of the set consisting of

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}.$$

Since $v_3 = 2v_1 + 2v_2$, the 3 vectors lie in a plane of $\mathbb{R}^3$ (the plane spanned by v_1 and v_2 , for example). However, the three vectors do not lie on the same line (they are not parallel), so $\text{rank}(v_1, v_2, v_3) = \dim(\text{plane}) = 2$.

Theorem

The rank of a set of vectors doesn't change if we add the linear combination of some of the vectors to a vector.

Matrices

Definition

A **matrix** is a rectangular array of numbers. An $m \times n$ (m -by- n) matrix has m rows and n columns.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

- the elements of A are a_{ij} , notation: $A = (a_{ij})$
- The set of the $m \times n$ real matrices is denoted by $\mathbb{R}^{m \times n}$, while the set of the $m \times n$ complex matrices is denoted by $\mathbb{C}^{m \times n}$.

Example.

$$\begin{pmatrix} -1 & 1 & 0 \\ 4 & -3 & -1 \end{pmatrix} \in \mathbb{R}^{2 \times 3}, \quad \begin{pmatrix} 2 - i \\ 3 + 2i \end{pmatrix} \in \mathbb{C}^{2 \times 1}$$

Matrices

Definition

- If $n = m$, then the matrix is a **square matrix**.
- The **main diagonal** of a matrix is the vector $(a_{11}, a_{22}, a_{33}, \dots, a_{kk})$, where $(k = \min\{m, n\})$.
- If A is a square matrix and all elements outside the main diagonal are zero, then we say that A is a **diagonal matrix**.
- If A is a square matrix and all elements **below or above** the main diagonal are zero, then we say that A is an **(upper or lower) triangular matrix**.
- The **identity matrix of size n** is the $n \times n$ matrix such that the elements on the main diagonal are equal to 1 and all other elements are zero. Notation: I_n .

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Matrix operations

1. Matrix addition

We can only add matrices of the same type.

Matrix addition: element-wise.

Example.

$$\begin{pmatrix} 1 & -4 \\ 0 & -2 \\ 3 & 3 \end{pmatrix} + \begin{pmatrix} 2 & -1 \\ 1 & 2 \\ 1 & 5 \end{pmatrix} = \begin{pmatrix} 3 & -5 \\ 1 & 0 \\ 4 & 8 \end{pmatrix}$$

2. Scalar multiplication

Scalar multiplication: element-wise.

Example.

$$2 \begin{pmatrix} 1 & 4 \\ 0 & 2 \\ 3 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 8 \\ 0 & 4 \\ 6 & 6 \end{pmatrix}$$

If A and B are row or column vectors, then the previous two operations are the vector addition and the scalar-vector multiplication.

Matrix operations

3. Matrix multiplication

Let $A = (a_{ij})$ be an $m \times k$ and $B = (b_{ij})$ be a $k \times n$ matrix. Then the product of A and B is the $m \times n$ matrix $C = (c_{ij})$, such that

$$c_{ij} = \sum_{r=1}^k a_{ir} b_{rj}.$$

The diagram illustrates the matrix multiplication $A \cdot B = C$ using matrix notation with dimensions indicated by brackets below:

- Matrix A is $m \times k$. It is shown as a matrix with rows and columns. The element a_{ik} is highlighted in a pink box.
- Matrix B is $k \times n$. It is shown as a matrix with rows and columns. The column containing elements $b_{1j}, \dots, b_{kj}$ is highlighted in a pink box.
- The resulting matrix C is $m \times n$. It is shown as a matrix with rows and columns. The element c_{ij} is highlighted in a pink box.

The equation is represented as:

$$\underbrace{\begin{pmatrix} * & \cdots & * & * \\ \vdots & & & \\ a_{i1} & \cdots & a_{ik} \\ \vdots & & & \\ * & \cdots & * & * \end{pmatrix}}_{m \times k} \underbrace{\begin{pmatrix} * & \cdots & b_{1j} & \cdots & * \\ \vdots & & \vdots & & \\ * & \cdots & \vdots & & * \\ \vdots & & \vdots & & \\ * & \cdots & b_{kj} & \cdots & * \end{pmatrix}}_{k \times n} = \underbrace{\begin{pmatrix} * & \cdots & * & \cdots & * & * \\ \vdots & & & & \\ * & \cdots & c_{ij} & \cdots & * & * \\ \vdots & & & & \\ * & \cdots & * & \cdots & * & * \end{pmatrix}}_{m \times n}$$

Properties of matrix multiplication

Example:

$$\underbrace{\begin{pmatrix} -1 & 2 \\ 1 & 0 \\ 3 & 2 \end{pmatrix}}_{3 \times 2} \underbrace{\begin{pmatrix} 4 & 1 \\ 2 & 5 \end{pmatrix}}_{2 \times 2} = \underbrace{\begin{pmatrix} 0 & 9 \\ 4 & 1 \\ 16 & 13 \end{pmatrix}}_{3 \times 2}$$

- Matrix multiplication is **not commutative**, that is, in general

$$AB \neq BA.$$

- If A is an $m \times n$ matrix, then

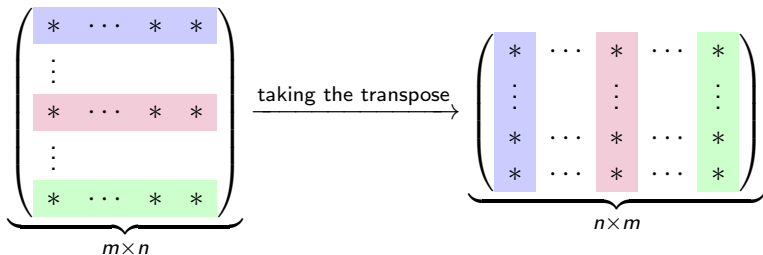
$$I_m \cdot A = A \quad \text{and} \quad A \cdot I_n = A.$$

Thus, multiplication by the identity matrix doesn't change the matrix.

Transpose of a matrix

Definition

Let A be an $m \times n$ matrix. The $n \times m$ matrix, which has rows as the columns of A is denoted by A^T and it is called the **transpose** of A .



Example:

$$A = \begin{pmatrix} 0 & -5 \\ -3 & 3 \\ -2 & 1 \end{pmatrix} \Rightarrow A^T = \begin{pmatrix} 0 & -3 & -2 \\ -5 & 3 & 1 \end{pmatrix}.$$

Symmetric matrices

Definition

Let A be a square matrix of order n . A is called **symmetric** if $A^T = A$.

Example: The matrix S below is symmetric.

$$S = \begin{pmatrix} 2 & -4 & 1 \\ -4 & 3 & 0 \\ 1 & 0 & 10 \end{pmatrix}$$

Exercise.

$$A = \begin{pmatrix} 1 & -4 \\ 3 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & -1 & 1 \\ 0 & -2 & 2 \end{pmatrix}, \quad u = \begin{pmatrix} 5 \\ 2 \end{pmatrix}, \quad v = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

Calculate the values of the following expressions, if they exist.

$$A^T, \quad B^T, \quad u^T, \quad v^T, \quad AB^T, \quad B^T A, \quad Au^T, \quad u^T A, \quad Bv, \\ u^T u, \quad uu^T, \quad v^T v, \quad vv^T$$

Determinants 1

Definition

Let $n \in \mathbb{N}$ and let σ denote a permutation of the set $\{1, 2, \dots, n\}$. (That is, $\sigma(i)$ denotes the i th element in the permutation.)

- We say that in the permutation σ the elements i and j are **in inversion** if $i < j$ but $\sigma(i) > \sigma(j)$.
- The permutation σ is called **even** if the number of pairs being in inversion in σ is even, and **odd** if this number is odd.

Examples. $\{1, 2, 3, 4\}$,

$$\sigma_1 = (1, 3, 4, 2)$$

Number of inversions: 2

$$\sigma_2 = (1, 2, 3, 4)$$

Number of inversions: 0

$$\sigma_3 = (4, 3, 2, 1)$$

Number of inversions: 6

$$\sigma_4 = (2, 3, 4, 1)$$

Number of inversions: 3

Determinants 2

Definition

Let $A = (a_{ij})$ be a square matrix. Let us choose n elements of A such that we choose exactly one element from each row and each column. The chosen elements:

$$a_{1\sigma(1)}, a_{2\sigma(2)}, \dots, a_{n\sigma(n)}.$$

The **determinant of A** is

$$\det(A) = |A| = \sum_{\sigma} \varepsilon(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \dots a_{n\sigma(n)}.$$

$$\text{Here } \varepsilon(\sigma) = \begin{cases} 1, & \text{if } \sigma \text{ is even,} \\ -1, & \text{if } \sigma \text{ is odd.} \end{cases}$$

There are $n!$ terms in the sum above.

Example.

① $n = 2$: $\det(A) = |A| = a_{11}a_{22} - a_{12}a_{21}.$

② $n = 3$: $\det(A) =$

$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}.$$

Properties of the determinant 1

Theorem

If A and B are square matrices of the same size, then

$$\det(AB) = \det(A) \cdot \det(B).$$

Properties

- The determinant of an identity matrix is 1. ($\det I_n = 1$)
- The determinant of a diagonal matrix is the product of the elements of the main diagonal.
- The determinant of a triangular matrix is the product of the elements in the main diagonal.

Examples: determinant of a diagonal and an upper triangular matrix

$$\det \begin{pmatrix} -2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{pmatrix} = (-2) \cdot 4 \cdot 3, \quad \det \begin{pmatrix} 5 & * & * \\ 0 & -3 & * \\ 0 & 0 & -1 \end{pmatrix} = 5 \cdot (-3) \cdot (-1)$$

Properties of the determinant 2

Properties

- $\det(A) = \det(A^T)$
- If A has a row full of zeros, then $\det(A) = 0$.
- If 2 rows of A are the same, then $\det(A) = 0$.
- If one row of A is a scalar multiple of another row, then $\det(A) = 0$.
- If a row of A is the linear combination of the other rows, then $\det(A) = 0$.

The properties above are true if we consider columns instead of rows.

Generally, if the rows or columns of A are linearly dependent vectors, then $\det(A) = 0$.

Properties of the determinant 3

Properties

- If we interchange 2 rows of A , then the sign of the determinant changes.
- If we multiply **one** row of A by a real number λ , then the obtained matrix has determinant $\lambda \cdot \det(A)$.
- If we multiply **each** row of A by a real number λ , then the obtained matrix has determinant $\lambda^n \cdot \det(A)$.
- The determinant doesn't change if we add a scalar multiple of a row to another row.

The properties above are true if we consider columns instead of rows.

Corollary

If $\det(A) \neq 0$, then the rows (or columns) of A are linearly independent vectors. Then if $A \in \mathbb{R}^{n \times n}$: its rows form a basis of $\mathbb{R}^n$.

How to calculate the determinant?

1. **The rule of Sarrus:** only for 2×2 and 3×3 determinants
for 2×2 matrices:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \implies \det(A) = a_{11}a_{22} - a_{12}a_{21}$$

for 3×3 matrices:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$\det(A) =$

$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}$$

Example.

Using the rule of Sarrus calculate the determinants of the following matrices.

$$\begin{pmatrix} -3 & 2 \\ 1 & 4 \end{pmatrix}, \quad \begin{pmatrix} -4 & -1 & -5 \\ 2 & 2 & 3 \\ -2 & 4 & 0 \end{pmatrix}$$

How to calculate the determinant?

2. **Gaussian elimination (or row reduction)**: using the following two modifications transform the matrix to triangular form.
- ▶ Interchanging two rows. (The determinant changes sign.)
 - ▶ Adding a scalar multiple of a row to another row. (The determinant does not change.)

The determinant of the original matrix is the product of the elements in the main diagonal of the triangular matrix, multiplied by $(-1)^k$, where k is the number of the performed row interchanges.

Example. Calculate $\det(A)$ using Gaussian elimination

$$A = \begin{pmatrix} -4 & -1 & -5 \\ 2 & 2 & 3 \\ -2 & 4 & 0 \end{pmatrix}$$

How to calculate the determinant?

3. **Laplace expansion:** We choose an arbitrary row or column of the matrix. E.g., if we choose the i^{th} row, then

$$\det(A) = |A| = \sum_{j=1}^n a_{ij} C_{ij}, \quad \text{where}$$

- ▶ C_{ij} is the cofactor of A corresponding to the element a_{ij} , that is,

$$C_{ij} = (-1)^{i+j} A_{ij},$$

- ▶ A_{ij} is the $(n-1) \times (n-1)$ determinant obtained from A by deleting the i^{th} row and j^{th} column of A .

Example. Calculate $\det(A)$ with the help of Laplace expansion.

$$A = \begin{pmatrix} -4 & -1 & -5 \\ 2 & 2 & 3 \\ -2 & 4 & 0 \end{pmatrix}$$

Rank of a matrix

The rank of a set of vectors: the dimension of the generated subspace.

Definition

The **rank of a matrix** is the rank of the system of column vectors of the matrix. Notation: $\text{rank}(A)$.

How to determine the rank of a matrix?

With Gaussian elimination.

The following operations do not change the rank of a matrix A :

- Interchanging 2 rows of A .
- Multiplying a row of A by a scalar $\lambda \neq 0$.
- Adding a scalar multiple of a row to another row.
- Discarding a row which is full of zeros.

Our goal: to obtain a triangular matrix by the operations above.

Rank: the number of the remaining rows.

Example

Determine the rank of the matrices below.

$$A = \begin{pmatrix} 3 & 5 & -6 \\ -1 & -2 & 1 \\ 2 & 6 & 5 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 & 0 \\ -2 & 2 & -6 \\ 0 & 4 & -4 \\ 3 & -3 & 9 \end{pmatrix}$$

$$\begin{aligned} A &= \begin{pmatrix} 3 & 5 & -6 \\ -1 & -2 & 1 \\ 2 & 6 & 5 \end{pmatrix} \rightarrow \begin{pmatrix} -1 & -2 & 1 \\ 3 & 5 & -6 \\ 2 & 6 & 5 \end{pmatrix} \rightarrow \begin{pmatrix} -1 & -2 & 1 \\ 0 & -1 & -3 \\ 0 & 2 & 7 \end{pmatrix} \\ &\rightarrow \begin{pmatrix} -1 & -2 & 1 \\ 0 & -1 & -3 \\ 0 & 0 & 1 \end{pmatrix} \implies \text{rank}(A) = 3 \end{aligned}$$

$$\begin{aligned} B &= \begin{pmatrix} 1 & 2 & 0 \\ -2 & 2 & -6 \\ 0 & 4 & -4 \\ 3 & -3 & 9 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 0 \\ 0 & 6 & -6 \\ 0 & 4 & -4 \\ 0 & -9 & 9 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 0 \\ 0 & 6 & -6 \end{pmatrix} \\ &\implies \text{rank}(B) = 2. \end{aligned}$$

The inverse of a matrix

Definition

A square matrix A of order n **has an inverse** or **invertible** if there exists a square matrix B of order n , such that

$$AB = BA = I_n.$$

Only a square matrix can have an inverse.

Not every square matrix has an inverse.

Theorem

If A has an inverse, then its inverse is uniquely determined. Notation: A^{-1} .

Example.

$$A = \begin{pmatrix} 4 & 3 \\ 7 & 5 \end{pmatrix} \qquad A^{-1} = \begin{pmatrix} -5 & 3 \\ 7 & -4 \end{pmatrix}$$

Properties of matrix inverse

- If A is invertible, then so is A^{-1} , and $(A^{-1})^{-1} = A$.
The inverse of the inverse matrix is the original matrix.

- If A and B are invertible matrices and AB exists, then $(AB)^{-1} = B^{-1}A^{-1}$.

The inverse of a product is the product of the inverses in reversed order.

- If A is invertible, then so is A^T , and $(A^{-1})^T = (A^T)^{-1}$.

The order of taking the inverse and the transpose can be swapped.

Determinant and matrix inverse

Definition

The square matrix A is called **regular** if $\det(A) \neq 0$.
Otherwise A is said to be **singular**.

Theorem

A matrix has an inverse if, and only if, it is regular (that is, its determinant is non-zero).

Proposition

If A is regular, then

$$\det(A)^{-1} = \det(A^{-1}).$$

With different notation,

$$\frac{1}{\det(A)} = \det(A^{-1}).$$

Inverse of a 2×2 matrix

If the 2×2 matrix

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

is regular, then its inverse is

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

Example:

$$A = \begin{pmatrix} 5 & 1 \\ 3 & 1 \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -3 & 5 \end{pmatrix}$$

Systems of linear equations

Example

A manufacturer produces two kinds of toys: wooden cars and wooden trains. The amounts of two raw materials needed to produce 1 piece of them are given in the table.

	car	train
wood	2	3
paint	5	4

Find the number of cars and trains produced on a given day, when we know that for the production there were used 540 units of wood and 1070 units of paint.

x_1 : number of cars

x_2 : number of trains

$$2x_1 + 3x_2 = 540$$

$$5x_1 + 4x_2 = 1070$$

$$2x_1 + 3x_2 = 540$$

$$5x_1 + 4x_2 = 1070$$

i.e.

$$\begin{pmatrix} 2 \\ 5 \end{pmatrix} x_1 + \begin{pmatrix} 3 \\ 4 \end{pmatrix} x_2 = \begin{pmatrix} 540 \\ 1070 \end{pmatrix}$$

$\begin{pmatrix} 540 \\ 1070 \end{pmatrix}$: the vector of the raw materials

$\begin{pmatrix} 2 \\ 5 \end{pmatrix}$: the vector of the raw materials needed for one car

$\begin{pmatrix} 3 \\ 4 \end{pmatrix}$: the vector of the raw materials needed for one train

Can you obtain the vector on the right-hand side as the linear combination of the vectors on the left-hand side?

If so, then in how many different ways?

The system

$$\begin{pmatrix} 2 \\ 5 \end{pmatrix} x_1 + \begin{pmatrix} 3 \\ 4 \end{pmatrix} x_2 = \begin{pmatrix} 540 \\ 1070 \end{pmatrix}$$

in a different form:

$$\begin{pmatrix} 2 & 3 \\ 5 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 540 \\ 1070 \end{pmatrix}$$

It is the so-called matrix-vector form of the system of linear equations.

Systems of linear equations (SLE)

Definition

The system of equations

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots \qquad \qquad \qquad \vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

where the numbers a_{ij} ($i \in \{1, \dots, m\}, j \in \{1, \dots, n\}$) and b_k ($k \in \{1, \dots, m\}$) are known, the variables $x_1, \dots, x_n$ are unknown, is called a **system of linear equations**.

- a_{ij} : the **coefficients of the system of linear equations**
- b_k : the **constant terms**

The **coefficient matrix** and the **augmented matrix**:

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix} \quad \text{and} \quad A|b = \left(\begin{array}{ccc|c} a_{11} & \dots & a_{1n} & b_1 \\ a_{21} & \dots & a_{2n} & b_2 \\ \vdots & & \vdots & \vdots \\ a_{m1} & \dots & a_{mn} & b_m \end{array} \right)$$

The **right-hand side vector** and the **solution vector**

$$b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \quad \text{and} \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

The corresponding matrix-vector equation: $Ax = b$.

Solvability of systems of linear equations – concepts

Definition

The system of linear equations is

- **solvable** if there exists at least one solution, that is, an x such that $Ax = b$ holds;
 - ▶ **determined** if there is exactly 1 solution;
 - ▶ **undetermined** if there are more than 1 solutions;
- **inconsistent** if it doesn't have a solution.

Car-train example: a unique solution (**determined**)

$$\begin{pmatrix} 2 \\ 5 \end{pmatrix} x_1 + \begin{pmatrix} 3 \\ 4 \end{pmatrix} x_2 = \begin{pmatrix} 540 \\ 1070 \end{pmatrix}$$

$x_1 = 150$, $x_2 = 80$, i.e.

$$x = \begin{pmatrix} 150 \\ 80 \end{pmatrix}$$

The system

$$2x_1 + x_2 = 3$$

$$6x_1 + 3x_2 = 9$$

i.e. the system

$$\begin{pmatrix} 2 \\ 6 \end{pmatrix} x_1 + \begin{pmatrix} 1 \\ 3 \end{pmatrix} x_2 = \begin{pmatrix} 3 \\ 9 \end{pmatrix}$$

is **undetermined**. The vector

$$x = \begin{pmatrix} s \\ 3 - 2s \end{pmatrix}$$

is a solution for every $s \in \mathbb{R}$. E.g.

$$x = \begin{pmatrix} 0 \\ 3 \end{pmatrix}, \quad \text{if } s = 0$$

$$x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \text{if } s = 1$$

The system

$$2x_1 + x_2 = 3$$

$$6x_1 + 3x_2 = 10$$

i.e. the system

$$\begin{pmatrix} 2 \\ 6 \end{pmatrix} x_1 + \begin{pmatrix} 1 \\ 3 \end{pmatrix} x_2 = \begin{pmatrix} 3 \\ 10 \end{pmatrix}$$

is **inconsistent**.

Using the vectors $\begin{pmatrix} 2 \\ 6 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ we can obtain only those vectors, whose second element is 3 times the first element.

The vector $\begin{pmatrix} 3 \\ 10 \end{pmatrix}$ is not in the space generated by the two vectors.

Theorem on solvability

Solvability: An SLE can be solved if and only if the vector b can be expressed as a linear combination of the column vectors of A . That is, if b is in the subspace spanned by the column vectors of A .

Number of solutions: Assume the SLE is solvable.

- If b can be expressed uniquely (i.e., the column vectors of A are linearly independent), then there exists a unique solution.
- If the column vectors of A are linearly dependent, then b can be obtained in several different ways as a linear combination of the columns of A , so there are infinitely many solutions.

Condition on solvability

- A system of linear equation is solvable if, and only if $\text{rank}(A) = \text{rank}(A|b)$.
- If it is solvable and
 - ▶ $\text{rank}(A) = n$ (where n is the number of unknown parameters), then the system is determined,
 - ▶ if $\text{rank}(A) < n$, then undetermined.

Solving a system of linear equations with Gaussian elimination

The set of solutions of a system of linear equations does not change, if we

- multiply an equation by a nonzero constant;
- add a scalar multiple of an equation to another equation;
- interchange two equations;
- discard an equation which is a scalar multiple of another equation.

We eliminate the numbers under the main diagonal of the augmented matrix with the modifications above. The resulting system is easier to solve.

- If during the process we obtain a row like $(0 \dots 0 | \neq 0)$, then the system of linear equations has no solution.
- If at the end of the process there are n number of not identically 0 rows, then the system is determined (there is a unique solution), if fewer number of rows remains, then undetermined (infinitely many solutions). (Here n is the number of the unknown parameters.)

Example

Solve the following SLE with Gaussian elimination.

$$\begin{array}{rcl} 2x_1 + 2x_2 + 3x_3 & = & -4 \\ -4x_1 - x_2 - 5x_3 & = & 12 \\ -2x_1 + 4x_2 & = & 10 \end{array}$$

Let's do elimination on the augmented matrix:

$$\left(\begin{array}{ccc|c} 2 & 2 & 3 & -4 \\ -4 & -1 & -5 & 12 \\ -2 & 4 & 0 & 10 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 2 & 2 & 3 & -4 \\ 0 & 3 & 1 & 4 \\ 0 & 6 & 3 & 6 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 2 & 2 & 3 & -4 \\ 0 & 3 & 1 & 4 \\ 0 & 0 & 1 & -2 \end{array} \right)$$

Backward substitution:

$$\left(\begin{array}{ccc|c} 2 & 2 & 3 & -4 \\ 0 & 3 & 1 & 4 \\ 0 & 0 & 1 & -2 \end{array} \right) \Rightarrow \begin{array}{rcl} 2x_1 + 2x_2 + 3x_3 & = & -4 \\ 3x_2 + x_3 & = & 4 \\ x_3 & = & -2 \end{array} \begin{array}{l} \longrightarrow x_1 = -1 \\ \longrightarrow x_2 = 2 \\ \longrightarrow x_3 = -2 \end{array}$$

Example

Solve the following SLE with Gaussian elimination.

$$-2x_1 + 4x_2 - 8x_3 = 2$$

$$x_1 + 3x_2 - x_3 = 4$$

$$3x_1 - x_2 + 7x_3 = 2$$

Let's do elimination on the augmented matrix:

$$\begin{pmatrix} -2 & 4 & -8 & | & 2 \\ 1 & 3 & -1 & | & 4 \\ 3 & -1 & 7 & | & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3 & -1 & | & 4 \\ -2 & 4 & -8 & | & 2 \\ 3 & -1 & 7 & | & 2 \end{pmatrix} \\ \rightarrow \begin{pmatrix} 1 & 3 & -1 & | & 4 \\ 0 & 10 & -10 & | & 10 \\ 0 & -10 & 10 & | & -10 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3 & -1 & | & 4 \\ 0 & 1 & -1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

In the last step we discarded the 3rd row, because it was a scalar multiple of the 2nd row, also, we divided the 2nd row by 10.

Backward substitution:

$$\begin{pmatrix} 1 & 3 & -1 & | & 4 \\ 0 & 1 & -1 & | & 1 \end{pmatrix} \Rightarrow \begin{array}{rcl} x_1 + 3x_2 - x_3 & = & 4 \\ x_2 - x_3 & = & 1 \end{array} \Rightarrow \begin{array}{l} x_3 = t, \quad t \in \mathbb{R} \\ x_2 = 1 + t, \\ x_1 = 1 - 2t \end{array}$$

Example

Solve the following SLE with Gaussian elimination.

$$\begin{array}{rcrcrcrcrl} -x_1 & + & 3x_2 & - & 7x_3 & & = & -2 \\ 3x_1 & + & 2x_2 & - & x_3 & & = & 6 \\ -2x_1 & + & 2x_2 & - & 6x_3 & & = & -3 \end{array}$$

Let's do elimination on the augmented matrix:

$$\begin{aligned} & \left(\begin{array}{ccc|c} -1 & 3 & -7 & -2 \\ 3 & 2 & -1 & 6 \\ -2 & 2 & -6 & -3 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} -1 & 3 & -7 & -2 \\ 0 & 11 & -22 & 0 \\ 0 & -4 & 8 & 1 \end{array} \right) \\ & \longrightarrow \left(\begin{array}{ccc|c} -1 & 3 & -7 & -2 \\ 0 & 1 & -2 & 0 \\ 0 & -4 & 8 & 1 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} -1 & 3 & -7 & -2 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right) \end{aligned}$$

(In the second step we divided the 2nd row by 11.)

This SLE is inconsistent, it doesn't have a solution.

Solutions of a system of linear equations

Definition

A system of linear equations is **homogeneous** if $b = 0$, thus if the matrix equation has the form $Ax = 0$. Otherwise it's called **nonhomogeneous**.

$$-x_1 + 3x_2 - 7x_3 = 0$$

$$3x_1 + 2x_2 - x_3 = 0$$

$$-2x_1 + 2x_2 - 6x_3 = 0$$

Remark: 0 is a solution of any homogenous SLE (it is the **trivial solution**).

Solutions of a homogeneous SLE

The solutions of a **homogeneous** system of linear equations form a vector subspace of $\mathbb{R}^n$ with dimension $n - \text{rank}(A)$.

Solutions of a nonhomogeneous SLE

The set of solutions of a (solvable) nonhomogeneous SLE is of the form $H + x_0$, where H is the solution set of the corresponding homogeneous system of linear equation ($Ax = 0$), and x_0 is a particular solution of the nonhomogeneous system.

Matrix inverse

Simultaneous solution of systems

If we have to solve the systems $Ax = b$ and $Ax = c$, then the elimination can be performed simultaneously, starting from the matrix $(A|b \ c)$.

$$\begin{array}{rclcl} 2x_1 + 2x_2 + 3x_3 & = & -4 & 2x_1 + 2x_2 + 3x_3 & = & 0 \\ -4x_1 - x_2 - 5x_3 & = & 12 & -4x_1 - x_2 - 5x_3 & = & -3 \\ -2x_1 + 4x_2 & = & 10 & -2x_1 + 4x_2 & = & -6 \end{array}$$

Inverse calculated with Gaussian elimination

Let $A \in \mathbb{R}^{n \times n}$ be an invertible matrix and denote its inverse by X . The determination of the inverse means the simultaneous solution of n systems of linear equations.

$$(A|I) \rightarrow \cdots \rightarrow (I|X)$$

Example

Find the inverse of $A = \begin{pmatrix} 3 & 5 & -6 \\ -1 & -2 & 1 \\ 2 & 6 & 5 \end{pmatrix}$

$$\begin{aligned}
& \left(\begin{array}{ccc|ccc} 3 & 5 & -6 & 1 & 0 & 0 \\ -1 & -2 & 1 & 0 & 1 & 0 \\ 2 & 6 & 5 & 0 & 0 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} -1 & -2 & 1 & 0 & 1 & 0 \\ 3 & 5 & -6 & 1 & 0 & 0 \\ 2 & 6 & 5 & 0 & 0 & 1 \end{array} \right) \\
& \rightarrow \left(\begin{array}{ccc|ccc} -1 & -2 & 1 & 0 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 & 0 \\ 0 & 2 & 7 & 0 & 2 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} -1 & -2 & 1 & 0 & 1 & 0 \\ 0 & -1 & -3 & 1 & 3 & 0 \\ 0 & 0 & 1 & 2 & 8 & 1 \end{array} \right) \rightarrow \\
& \left(\begin{array}{ccc|ccc} -1 & -2 & 0 & -2 & -7 & -1 \\ 0 & -1 & 0 & 7 & 27 & 3 \\ 0 & 0 & 1 & 2 & 8 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} -1 & 0 & 0 & -16 & -61 & -7 \\ 0 & -1 & 0 & 7 & 27 & 3 \\ 0 & 0 & 1 & 2 & 8 & 1 \end{array} \right) \\
& \rightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 16 & 61 & 7 \\ 0 & 1 & 0 & -7 & -27 & -3 \\ 0 & 0 & 1 & 2 & 8 & 1 \end{array} \right) \Rightarrow A^{-1} = \left(\begin{array}{ccc} 16 & 61 & 7 \\ -7 & -27 & -3 \\ 2 & 8 & 1 \end{array} \right)
\end{aligned}$$

$\Rightarrow$ Gaussian elimination can be used to calculate the inverse of a matrix

Euclidean vector spaces

Definition

The **scalar product** (or **dot product** or **inner product**) of $x, y \in \mathbb{R}^n$ is defined as:

$$\langle x, y \rangle := \sum_{i=1}^n x_i y_i = x^T y$$

Example. Calculate the scalar product of x and y .

$$x = \begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix}, \quad y = \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}$$

$$\langle x, y \rangle = x^T y = \begin{pmatrix} -1 & 2 & -2 \end{pmatrix} \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} = -1 \cdot 3 + 2 \cdot 0 + (-2)(-1) = -1$$

Properties of the real scalar product:

1. **symmetry**: $\langle x, y \rangle = \langle y, x \rangle$, $\forall x, y \in \mathbb{R}^n$
2. **additivity in both arguments**: for all $x, y \in \mathbb{R}^n$

$$\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle \quad \text{and} \quad \langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$$

3. **homogeneity in both arguments**: for all $x, y \in \mathbb{R}^n, \lambda \in \mathbb{R}$

$$\langle \lambda x, y \rangle = \lambda \langle x, y \rangle \quad \text{and} \quad \langle x, \lambda y \rangle = \lambda \langle x, y \rangle$$

4. **positive definiteness**: $\langle x, x \rangle \geq 0$, $\forall x \in \mathbb{R}^n$ and $\langle x, x \rangle = 0 \Leftrightarrow x = 0$

Norm in $\mathbb{R}^n$

Definition

The (2-, or Euclidean) **norm** of $x \in \mathbb{R}^n$ is

$$\|x\| := \sqrt{\langle x, x \rangle} = \sqrt{\sum_{i=1}^n x_i^2}$$

Properties of the norm:

1. **nonnegativity**: $\|x\| \geq 0$, $\forall x \in \mathbb{R}^n$, moreover $\|x\| = 0 \Leftrightarrow x = 0$;
2. **absolute homogeneity**: $\|\lambda x\| = |\lambda| \|x\|$, $\forall x \in \mathbb{R}^n, \lambda \in \mathbb{R}$
3. **triangle inequality**: $\|x + y\| \leq \|x\| + \|y\|$, $\forall x, y \in \mathbb{R}^n$

Example. Determine the norms of the previous x and y vectors.

$$\|x\| = \sqrt{(-1)^2 + 2^2 + (-2)^2} = \sqrt{9} = 3$$

$$\|y\| = \sqrt{3^2 + 0^2 + (-1)^2} = \sqrt{10}$$

Euclidean vector space

The space $\mathbb{R}^n$ with the previous scalar product is called real **Euclidean vector space**.

Cauchy-Schwarz inequality

For all vectors x, y of the real Euclidean vector space

$$|\langle x, y \rangle| \leq \|x\| \cdot \|y\|.$$

Equality holds if, and only if, one vector is a scalar multiple of the other.

Angle between two real vectors

If $x, y \in \mathbb{R}^n$ and $x, y \neq 0$, then

$$-1 \leq \frac{\langle x, y \rangle}{\|x\| \cdot \|y\|} \leq 1.$$

The angle between the real $x, y \neq 0$ vectors is the angle $\alpha \in [0, \pi]$, for which

$$\cos \alpha = \frac{\langle x, y \rangle}{\|x\| \cdot \|y\|}.$$

Thus if $x, y \neq 0$, then their angle is $\alpha \in [0, \pi]$, if

$$\cos \alpha = \frac{\langle x, y \rangle}{\|x\| \cdot \|y\|}$$

Remark

The angle between the real vectors $x, y \neq 0$ is $\frac{\pi}{2}$ (the vectors are perpendicular), if and only if $\langle x, y \rangle = 0$.

Example. Calculate the angle between the given vectors.

(a)

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

(b)

$$\begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

(c)

$$\begin{pmatrix} 3 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} \sqrt{3} \\ 1 \end{pmatrix}$$

(d)

$$\begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$$

(e)

$$\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 5 \\ 1 \\ 3 \end{pmatrix}$$

Definition

The vector $x \in \mathbb{R}^n$ is a **unit vector** if $\|x\| = 1$.

For an arbitrary nonzero vector x the vector $\frac{x}{\|x\|}$ is a unit vector.

Definition

- The vectors $x, y \in \mathbb{R}^n$ are called perpendicular, or **orthogonal**, if $\langle x, y \rangle = 0$.
- The vectors $v_1, \dots, v_m$ are called **orthogonal**, if they are pairwise orthogonal.
- They are called **orthonormal**, if they are orthogonal and are also unit vectors: $\|v_i\| = 1, i = 1, \dots, m$.

Theorem

Orthogonal vectors are linearly independent.

Example

Find vectors in $\mathbb{R}^3$ which are orthogonal to

$$x = \begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix}$$

Solution. We are looking for a vector $y = (y_1, y_2, y_3)^T$ such that $\langle x, y \rangle = 0$:

$$-2y_1 + 3y_2 + 4y_3 = 0.$$

This is a (homogeneous) system of linear equations.

Solution:

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = s \cdot \begin{pmatrix} \frac{3}{2} \\ 1 \\ 0 \end{pmatrix} + t \cdot \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \quad s, t \in \mathbb{R}.$$

The solution space is a plane through the origin.

Example

Find vectors in $\mathbb{R}^3$ which are orthogonal to the vectors

$$x = \begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix} \quad \text{and} \quad y = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$$

Solution. We are looking for a vector $z = (z_1, z_2, z_3)^T$ such that $\langle x, z \rangle = 0$ and $\langle y, z \rangle = 0$:

$$\begin{aligned} -2z_1 + 3z_2 + 4z_3 &= 0 \\ 2z_1 \quad \quad + 2z_3 &= 0. \end{aligned}$$

This is again a homogeneous system of linear equations.

$$\left(\begin{array}{ccc|c} -2 & 3 & 4 & 0 \\ 2 & 0 & 2 & 0 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} -2 & 3 & 4 & 0 \\ 0 & 3 & 6 & 0 \end{array} \right) \Rightarrow \begin{aligned} -2z_1 + 3z_2 + 4z_3 &= 0 \\ z_2 + 2z_3 &= 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow \quad z_3 &= t, \quad t \in \mathbb{R} \\ z_2 &= -2t, \\ z_1 &= -t \end{aligned} \quad \Rightarrow \quad \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = t \cdot \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

Example

Find the value of α , if x and y are orthogonal.

$$x = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \quad \text{and} \quad y = \begin{pmatrix} 6 \\ \alpha \end{pmatrix}$$

The vectors x and y are orthogonal if, and only if, their inner product is 0.
Their inner product:

$$\langle x, y \rangle = -2 \cdot 6 + 3 \cdot \alpha = -12 + 3\alpha.$$

Thus

$$-12 + 3\alpha = 0$$

$$3\alpha = 12$$

$$\alpha = 4.$$

Linear maps

When $A \in \mathbb{R}^{m \times n}$, then A maps every column vector of size n into a column vector of size m .

Example:

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} \in \mathbb{R}^{2 \times 3}$$

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -4 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 - x_3 \\ x_2 - x_3 \end{pmatrix}$$

Example:

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ -x_3 \end{pmatrix}$$